MAgent Theory 2.0

A Physicist's Companion

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Chapter 1

What an Agent Carries

A classical statistical description distinguishes a phase-space point from a distribution over phase space. The same distinction begins this theory: $k \in \mathsf{K}$ is a represented possibility, while q is a probability law over such possibilities. The agent carries the law and mechanisms for changing it. Its contents need not be positions and momenta, and the statistical law need not describe an ensemble of physically separate agents.

1.1 Beliefs, models, and predictions

There are two sectors. The belief law q_i describes state contents k. The model law s_i describes model presentations $m \in \mathsf{M}$. Their most general joint description is

$$\zeta_i(dk, dm) = q_i(dk \mid m) s_i(dm). \tag{1.1}$$

The notation $q_i(dk \mid m)$ means a conditional law, and $q_i(dk)$ without the condition denotes its marginal. Choosing $\zeta_i = q_i \otimes s_i$ removes dependence between the sectors in that description. It still permits coupled equations for q_i and s_i, just as coupled equations for two coordinates do not require a statistical ensemble over those coordinates.

A model presentation becomes predictive through a normalized evaluator $G_i(m, dk)$. For each m, this assigns a probability law to the state contents. Averaging the evaluator over the model law gives

$$(s_i G_i)(A) = \int G_i(m, A) s_i(dm). \tag{1.2}$$

For example, let k, m be binary, with $s_i(m = 1) = 0.6$, $G_i(1, \{1\}) = 0.9$, and $G_i(0, \{1\}) = 0.2$. The model mixture assigns probability 0.62 to $k = 1$. A present belief $q_i(k = 1) = 0.8$ is a different object. An interaction rule can bring these descriptions into agreement, revise the model law, or change both. Those dynamics must be specified.

An evaluator is a Markov kernel: a function whose output is a probability measure. Ordinary functions are included as the case in which the output measure is concentrated at one value. This language accommodates discrete, continuous, singular, and multimodal laws without selecting a Gaussian family. When differentiating laws later, we will restrict to a regular statistical family or positive finite tables.

1.2 A belief is not a generative model

The evaluator is half of a generative model. The other half is a record likelihood $\ell_i(k, m; do)$, which says how represented contents would show up in the agent's own registers. Together with a prior

$r_i(dm)$ over presentations they form one normalized joint law,

$$P_i(dm, dk, do) = r_i(dm)G_i(m, dk)\ell_i(k, m; do). \tag{1.3}$$

Each presentation m labels a complete forward model: a prior over contents and a likelihood for records. The agent's laws q_i and s_i are something else. They are its current beliefs, the recognition law that approximates the posterior $P_i(dm, dk \mid o)$ once a record o is in hand, and their quality is scored by the variational free energy $D_{\mathrm{KL}}(\zeta_i \| P_i(\cdot \mid o)) - \log P_i(o)$. The analogy is a detector with a calibration curve. The calibration curve is the generative model; the current estimate of the signal is the belief. Two detectors of one design share the calibration curve and can still hold different estimates; two detectors with the same estimate predict the same counts only if their calibrations also agree.

The same distinction organizes populations. Agents whose generative models agree up to a relabeling of presentations and contents form a species. Sharing a model is a fact about mechanisms. Agreement of beliefs is a fact about states, and a record registered by one member changes its beliefs without changing those of a member that did not register it.

Without records the best belief under a model is the prior itself, with the conditional $q_i(dk \mid m)$ equal to $G_i(m, dk)$. Under the product restriction $\zeta_i = q_i \otimes s_i$ this is no longer reachable: the best belief at fixed s_i is the geometric average $q_i \propto \exp \int \log g_i(m, \cdot) s_i(dm)$ of the evaluator densities, not the arithmetic mixture $s_i G_i$. The product description therefore already separates what an agent believes from what its models predict.

1.3 A record has an address inside the agent

Let $x_i \in \mathsf{S}_i$ denote the complete internal state, including its laws, interfaces, and any memory. A record readout is an ordinary measurable function

$$\rho_i : \mathsf{S}_i \longrightarrow \mathsf{O}_i. \tag{1.4}$$

The output alphabet O_i specifies the distinctions the agent can read. A binary register gives $\mathsf{O}_i = \{0, 1\}$; a real-valued comparison can use $\mathbb{R}^r$. The detector analogy is precise at this level: an internal state contains a readable distinction. It does not require an outside observer to inspect it. A detector reading must also not depend on the coordinates in which it is written down. A readout of the first coordinate of a displayed mean would change under a mere rotation of the agent's frame, so readouts are required to be invariant, or to transform by a declared rule, under such passive changes.

Write $x \in \mathsf{S}$ for the complete population configuration. A deterministic interaction $F_{ij} : \mathsf{S} \to \mathsf{S}$ produces the record

$$o_{ij} = \rho_i\big(\mathrm{pr}_i\, F_{ij}(x)\big), \tag{1.5}$$

where pr_i selects the receiver's state. For a stochastic transition $\Gamma_{ij}(x, dx')$, the probability of a record in A is

$$\mathcal{R}_{ij}(x, A) = \Gamma_{ij}\big(x, \{x' : \rho_i(\mathrm{pr}_i\, x') \in A\}\big). \tag{1.6}$$

This is a pushforward: collect the probability of every next state that displays an outcome in A. The transition specifies what happens; the readout specifies what is accessible. A persistent record requires that some later readout still recover the distinction. A transient comparison and a memory trace therefore have different dynamical requirements.

A likelihood on represented contents has another role. For instance, the expression $o = A_i k_i - B_{ij} k_j + \epsilon$ gives a comparison value once the contents and noise are given. Their joint law predicts possible comparisons. A registration rule must still place the outcome in an internal state and specify the response to it.

1.4 Which probability space is being used?

The configuration x contains the agents' laws. A tuple of represented contents $(k_i, m_i)_i$ belongs to a different space. A probability law over possible configurations is a third construction, useful when a preparation or stochastic dynamics has been declared. Deterministic motion of x requires neither sampling the contents nor drawing an ensemble of configurations at each step.

This distinction also clarifies correlations. Two independent fair bits have probability $1/4$ for each pair. Condition on a shared record that the bits agree. Their marginals remain fair, but the conditional joint law has probability $1/2$ on each of $(0, 0)$ and $(1, 1)$. A calculation of the next joint comparison needs this dependence; the marginals do not contain it. A coupled product-family evolution remains a possible model, but it will not represent that exact conditional joint law. It does not even keep the fair marginals. The best product description of the hard agreement record is a point mass on $(0, 0)$ or on $(1, 1)$. For a soft record $\propto e^{J\sigma_1\sigma_2}$ in spin variables, the symmetric product description is the mean-field magnet $m = \tanh(Jm)$: unmagnetized for $J \leq 1$ and spontaneously magnetized above $J = 1$, although the exact posterior keeps zero magnetization for every J. Whether a correlation is needed is decided by the response being predicted.

A third kind of law joins the contents of several agents, or of one agent at several contexts, into one joint law of fields. Only such a law can predict a record that compares the same sample along two routes. A comparison of the two route laws cannot see a loop that leaves the transported law unchanged, such as a rotation of an isotropic Gaussian, whereas the joint law of a sample and its image along the loop determines the loop map.

The laws can be attached to local domains of a context base $\mathcal{C}$. This is the section picture: a context selects the law applicable there, and a frame specifies its coordinates. Most interactions below occur at a common context. Moving between contexts introduces the connection problem in chapter 3.

Chapter 2

Frames, Messages, and Interactions

An agent's probability law and the coordinates used to describe that law are different objects. Communication introduces a third object: a rule for translating, filtering, or modifying what another agent supplies. Keeping these distinctions explicit prevents a coordinate change from being mistaken for an interaction, or a selective message from being mistaken for an invertible transport.

2.1 Reading a link from source to receiver

There are two sectors. Write $\nu_i^q = q_i$ for the belief marginal and $\nu_i^s = s_i$ for the model marginal. In a finite-dimensional realization, a sector $a \in \{q, s\}$ has a reference vector space V_a, an agent space E_i^a, and an invertible presentation

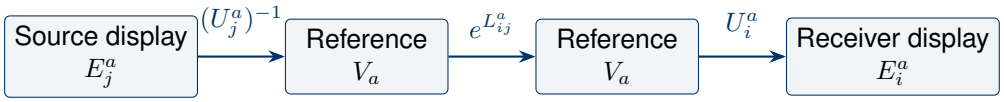
$$U_i^a : V_a \longrightarrow E_i^a, \qquad \bar{\nu}_i^a = (U_i^a)_{\#}^{-1} \nu_i^a. \tag{2.1}$$

The pushforward notation expresses the represented possibilities in reference coordinates. The space E_i^a is agent i's displayed copy of V_a: at one context all agents' laws lie in one fiber of the statistical bundle, and the presentations are frames used to display them. The two sectors may use representations of the same frame group while having different dimensions, laws, and interaction rules. Whether a passive change is one group element acting on both sectors, or independent elements for each sector, is a declared choice, and the collective geometry of chapter 3 depends on it.

A law-dependent reference generator and its displayed link are

$$L_{ij}^a = \mathcal{L}_{ij}^a(\bar{\nu}_i^a, \bar{\nu}_j^a), \qquad \Theta_{ij}^a = U_i^a e^{L_{ij}^a} (U_j^a)^{-1}. \tag{2.2}$$

The index order is receiver first, source second. Reading from right to left, the source is expressed in reference coordinates, the generator acts there, and the result is expressed at the receiver. The function $\mathcal{L}_{ij}^a$ is a constitutive choice. The exponential parameterization does not derive it or equate the belief and model generators. It is also only one kind of interaction. The general interaction is a normalized Markov channel, possibly noisy or information-discarding; the exponential link is the special deterministic channel $y \mapsto \Theta_{ij}^a y$. Its values e^L lie in $\exp(\mathfrak{gl}(V_a))$, which in dimension at least two misses some determinant-one matrices, such as $\begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}$, and every reflection [9].



$$\Theta_{ij}^a = U_i^a e^{L_{ij}^a} (U_j^a)^{-1}$$

$a = q$ or s: two separately specified interactions

Figure 2.1: The source presentation is undone by $(U_j^a)^{-1}$, the interaction $e^{L_{ij}^a}$ acts within the reference space, and U_i^a presents the result at the receiver. Endpoint frames and the intervening interaction have separate roles.

2.2 Changing coordinates

For a passive change $y_i' = g_i^a y_i$, both the law and its presentation change: $\nu_i^{a\prime} = (g_i^a)_\# \nu_i^a$ and $U_i^{a\prime} = g_i^a U_i^a$. The reference law stays fixed, giving

$$\Theta_{ij}^{a\prime} = g_i^a \Theta_{ij}^a (g_j^a)^{-1}. \tag{2.3}$$

This endpoint covariance is familiar from gauge link variables [32], here in display coordinates. It holds for every generator rule, because the reference laws do not change, and so it constrains nothing: the frames act as a dressing field [11]. Every interaction cost can be computed entirely in reference coordinates. Metrics and additional model data must also be re-expressed when forming scalar predictions. Covariance alone supplies no physical symmetry or evolution law.

Changing the common reference presentation is a separate operation. Under $U_i^a \mapsto U_i^a h_a$, the reference laws become $(h_a^{-1})_\# \bar{\nu}_i^a$. Reference consistency requires the constitutive rule, including any interface data, to give $L_{ij}^{a\prime} = h_a^{-1} L_{ij}^a h_a$. The identity $e^{h_a^{-1} L h_a} = h_a^{-1} e^L h_a$ then leaves the displayed link unchanged. This is the substantive symmetry requirement. It is not automatic, and the main text imposes it as an explicit standing hypothesis wherever a conclusion should not depend on the choice of reference frame.

2.3 Selective interfaces and an explicit pair

Sending and receiving can be represented by

$$T_j^a : V_a \longrightarrow H_a, \qquad O_i^a : H_a \longrightarrow V_a, \qquad L_{ij}^a = O_i^a T_j^a. \tag{2.4}$$

The sender T_j^a extracts a message; the receiver O_i^a inserts its effect. A direct content map $O_i^a T_j^a$ can lose dimensions. Its exponential is always invertible and acts as identity on $\ker T_j^a$. Thus exponentiating a low-rank generator retains unread directions, and the link compares complete laws. Sending only the extracted message is a different, information-discarding operation. General normalized Markov kernels allow communication without any linear realization.

One explicit law-dependent choice uses finite second moments and positive-definite covariance. Let μ_i^a, Σ_i^a be the reference mean and covariance. On nonself edges choose a scalar λ_{ij}^a and set

$$L_{ij}^a = \lambda_{ij}^a \mu_i^a (\mu_j^a)^T (\Sigma_j^a)^{-1}. \tag{2.5}$$

The row $(\mu_j^a)^T(\Sigma_j^a)^{-1}$ sends a scalar message; the column $\lambda_{ij}^a\mu_i^a$ receives it. Under a reference change, the mean transforms by h_a^{-1} and covariance by congruence. Their product consequently transforms by conjugation. These assumptions concern moments, not Gaussian shape. For these links the primitive self link is set to the identity separately. In general the self channel is constitutive data, and some later constructions use a nonidentity one.

For a concrete non-Gaussian example, let Z have independent centered Laplace coordinates of variance one in two dimensions. Give two agents the laws of $e_1 + Z$ and $e_2 + Z$, with identity frames and unit coefficients. Then $\Sigma_1 = \Sigma_2 = I$, $L_{12} = E_{12}$, and $L_{21} = E_{21}$. Since each generator squares to zero,

$$\Theta_{12}\Theta_{21} = (I + E_{12})(I + E_{21}) = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}. \tag{2.6}$$

Here E_{ab} has its only nonzero entry, one, in row a, column b. The directed communication loop is nontrivial. It establishes neither a connection over contexts nor a physical curvature. Writing $\Theta_{12} = \Omega_{12}N_2^{1/2}$ with $N_2 = \Theta_{21}\Theta_{12}$, the reciprocal part Ω_{12} is here a rotation with $\Omega_{12}\Omega_{21} = I$, so this two-cycle measures nonreciprocity, not holonomy.

A further limit concerns what any cost of the form $D_{\mathrm{KL}}(\nu_i \| \nu_j \mathcal{K}_{ij})$ can see. A loop map H enters only through the transported law $H_{\#}\nu$. If H fixes the law, as a rotation fixes a centered isotropic Gaussian, the cost cannot register it. The joint law of a single sample Y and its image HY does determine H almost everywhere, so detecting such a loop requires a record that compares one sample along two routes.

Chapter 3

From Statistical Change to Base Geometry

The agent graph and the context base describe different directions of comparison. An edge $j \to i$ compares agents at one context. A path in the base $\mathcal{C}$ compares states at successive contexts. A product of communication links can be nontrivial without defining curvature on $\mathcal{C}$. Conversely, one agent can carry a curved connection over contexts; collective averaging is not required. In the dynamics of the main text no update reads a connection, so everything in this chapter is a way of describing evolved or supplied sections, not a force acting on them.

3.1 Relating neighboring contexts

To describe such a connection, suppress the sector label and fix one agent. On a coordinate neighborhood, take a twice continuously differentiable presentation $U(c)$ and a supplied, continuously differentiable matrix-valued one-form $a = a_\mu(c)dc^\mu$. Nothing below derives a from the agents' laws or links. Declare directional links whose first-order behavior is

$$\Theta(c + \delta c, c) = U(c + \delta c)e^{-a_\mu(c)\delta c^\mu}U(c)^{-1} + O(|\delta c|^2). \tag{3.1}$$

The displacement enters linearly at first order, and zero displacement gives identity. These are additional properties beyond having a finite interaction matrix at each context. For a principal G-connection, the generators must belong to the represented Lie algebra; unrestricted matrices instead define general linear bundle connections.

Using $D = d + B$, expansion identifies

$$B = UaU^{-1} - dUU^{-1}. \tag{3.2}$$

The first term carries the declared directional interaction into displayed coordinates. The second compensates for the change of presentation along the path. Their relationship is exactly the operator identity

$$d + B = U(d + a)U^{-1}. \tag{3.3}$$

Indeed, acting on a vector field f gives $Ud(U^{-1}f) = df - dUU^{-1}f$; the remaining term is $UaU^{-1}f$. Squaring the operator therefore gives

$$F = dB + B \wedge B = U(da + a \wedge a)U^{-1}. \tag{3.4}$$

This is the mathematical connection to gauge theory, with the sign convention fixed by the displayed covariant derivative [31].

3.2 Curvature and paths

Curvature need not come from spatially varying coefficients. On a two-dimensional context patch choose $U = I$, $a_x = E_{12}$, and $a_y = E_{21}$. Both directional generators have rank one, but

$$F_{xy} = [E_{12}, E_{21}] = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{3.5}$$

The matrix E_{ab} has one unit entry in position (a, b). Applying the two infinitesimal operations in opposite orders gives different results. This example supplies directional data; it does not derive them from the agents' laws. A presentation change alone, $B = -dUU^{-1}$, instead has zero curvature.

Finite path transport solves

$$\frac{dT}{dt} = -B(\dot{\gamma}(t))T(t), \qquad T(0) = I. \tag{3.6}$$

Its solution is a path-ordered exponential. Concatenating paths composes their transports; reversing the same path gives the inverse. Two independently specified attending edges can have unrelated generators, so their opposite labels do not impose this inverse relation. The distinction is necessary when borrowing Wilson-loop language from gauge theory [32].

3.3 Local descriptions and global compatibility

Endpoint covariance requires

$$\Theta'(c_2, c_1) = g(c_2)\Theta(c_2, c_1)g(c_1)^{-1}. \tag{3.7}$$

Differentiating the endpoints produces the affine law

$$B' = gBg^{-1} - dgg^{-1}. \tag{3.8}$$

Likewise, keeping the intrinsic transport fixed under $U' = Uh$ requires $a' = h^{-1}ah + h^{-1}dh$. A directional receiving–sending product that transforms only by conjugation is a tensorial one-form. It can supply the difference between two connections, but does not by itself supply the full affine transformation law.

Global compatibility adds another requirement. If patch coordinates obey $v_\alpha = g_{\alpha\beta}v_\beta$, then $g_{\alpha\beta}g_{\beta\gamma} = g_{\alpha\gamma}$ on triple overlaps, and local connection forms must satisfy the affine law there. A nontrivial bundle may have no single global presentation. Even a trivial bundle can carry flat transport with global memory: on a circle, the scalar connection $B = bd\theta$ has $F = 0$ but holonomy $e^{-2\pi b}$ around one turn. For real $b \neq 0$, this differs from identity.

3.4 Distinguishability supplies a fiber metric

A connection tells us how to compare laws at neighboring contexts. To measure the change remaining after that comparison, use the Fisher metric on a declared regular statistical family. Let ζ_θ be a family of probability laws, with a centered square-integrable score ℓ_v in parameter direction v. In a dominated differentiable realization,

$$\ell_v = \left.\frac{d}{d\epsilon}\right|_{\epsilon=0} \log p_{\theta+\epsilon v}, \qquad g^F_\zeta(v,w) = \mathbb{E}_\zeta[\ell_v \ell_w]. \tag{3.9}$$

Quadratic-mean differentiability supplies a more general score definition. Positive definiteness requires that the admitted tangent directions be statistically distinguishable. These hypotheses apply to the chosen family, not to every element of the measurable space of laws. The laws of the family may have different null sets [2]; a finite-dimensional family is still dominated by a countable mixture of its members. Up to a constant factor, the Fisher metric is the only metric on finite statistical models invariant under Markov congruent embeddings [6], which is the reason for choosing it.

The relevant law is the full $\zeta(dk, dm) = q(dk \mid m)s(dm)$. Its log density has a marginal part and a conditional part. The conditional score averages to zero for each m, so the cross term disappears:

$$g^F_\zeta = g^F_s + \int g^F_{q(\cdot|m)} s(dm). \tag{3.10}$$

For the product family this is $g^F_q + g^F_s$. For a general joint family the second term is conditional geometry. It cannot be replaced by the geometry of the state marginal.

For example, take $k, m \in \{-1, 1\}$ and

$$\zeta_\rho(k,m) = \frac{1+\rho km}{4}, \qquad |\rho| < 1. \tag{3.11}$$

Both marginals are uniform for every ρ. The score is $km/(1+\rho km)$, giving

$$g^F_\zeta(\partial_\rho, \partial_\rho) = \frac{1}{2}\left(\frac{1}{1+\rho} + \frac{1}{1-\rho}\right) = \frac{1}{1-\rho^2}. \tag{3.12}$$

The law changes through dependence even though neither marginal changes. This is why the statistical metric must follow the actual probabilistic description.

3.5 Pulling distinguishability onto the context base

Let $\zeta(c)$ be an admitted smooth section and let ω be its connection. Subtract the horizontal motion from the derivative of the section:

$$D^\omega \zeta = \mathrm{ver}^\omega\, T\zeta, \qquad h^\omega_\zeta(X,Y) = g^F_\zeta(D^\omega \zeta X, D^\omega \zeta Y). \tag{3.13}$$

The tangent vectors X, Y now lie in the context base. The tensor h^ω_ζ measures how much the represented law changes when the context changes, after the declared transport accounts for the

change of frame. In a local chart it uses $D^A\zeta = d\zeta + J_\zeta A$, where $J_\zeta\xi$ is the infinitesimal action of the Lie algebra element ξ on the law.

Under second-order relative-entropy regularity, move a small distance ϵX, transport the neighboring law back, and compare it with the original:

$$D_{\mathrm{KL}}\left(\zeta(c)\middle\|T^{\omega}_{c+\epsilon X\to c\#}\zeta(c+\epsilon X)\right) = \frac{\epsilon^2}{2}h^{\omega}_{\zeta}(X,X) + o(\epsilon^2). \tag{3.14}$$

The familiar local quadratic behavior of KL therefore measures contextual distinguishability. The transport is part of the comparison.

For a simple case with identity transport, let the law at real context c be $\mathrm{Ber}(p(c))$, with $0 < p(c) < 1$. Direct differentiation of its two log probabilities gives

$$h = \frac{p'(c)^2}{p(c)(1-p(c))}dc^2. \tag{3.15}$$

The same change of success probability is more distinguishable near a deterministic outcome. If $p'(c) = 0$, this family records no first-order distinction in that context direction. The example uses no Gaussian assumption.

A passive gauge change transforms both the section and connection, leaving the tensor unchanged. Choosing a different connection changes what is subtracted. If A changes by a Lie-algebra-valued one-form α, the statistical derivative changes by $J_\zeta\alpha$; its squared Fisher norm acquires both cross terms and the squared norm of that added vector. Gauge invariance is thus a statement about presentation, not a claim that all transport rules give the same geometry.

Positive-definite fiber geometry makes $\ker h^{\omega}_{\zeta} = \ker D^{\omega}\zeta$. Constant rank gives a metric on the vector spaces obtained by quotienting these null directions. To obtain a lower-dimensional context manifold, the null directions must also fit together into leaves and the metric must descend along them. Both requirements are measured by the curvature F of the connection: for null directions X, Y, the derivative of the section along $[X, Y]$ is $-J_\zeta F(X, Y)$, so the null directions integrate exactly when $J_\zeta F$ vanishes on them, and a similar curvature expression decides whether the metric descends. A flat connection always passes both tests. The supplied connection $dz - xdy$ for a translation family illustrates failure: its square has rank one everywhere, but its kernel is not integrable because $(dz - xdy) \wedge d(dz - xdy)$ is nonzero. Counting statistical directions does not by itself construct a quotient space of contexts.

3.6 Choosing a collective connection

Suppose several laws have been placed in compatible fibers by declared comparison maps. Give their product tangent space the positive metric

$$\mathsf{G} = \sum_i w_i g^F_{p_i}, \qquad w_i > 0. \tag{3.16}$$

The weights specify how the individual statistical changes are compared. A common infinitesimal frame motion ξ changes the tuple by $J_{\boldsymbol{p}}\xi$. For each contextual direction X, select the correction a

that minimizes

$$\|d\boldsymbol{p}(X) + J_{\boldsymbol{p}}a\|^2_{\mathsf{G}}. \tag{3.17}$$

Differentiation with respect to a gives the normal equation

$$\mathsf{G}(JA^F(X), J\xi) = -\mathsf{G}(d\boldsymbol{p}(X), J\xi) \quad \text{for every } \xi. \tag{3.18}$$

The residual is orthogonal to all common frame motions. This is the mechanical connection of Smale [28], used to separate rotation from shape change in molecules [14] and the n-body problem [22], with the Fisher metric as the kinetic metric. The residual is the shape change: under a free proper action, $h^{\min}$ is the pullback of the quotient metric on the space of orbits [24]. Its null directions are those along which only the orbit moves, and they always integrate.

For an explicit calculation, take two unit-variance normal location laws with means $\mu_1 = m + d$ and $\mu_2 = m - d$, and give them equal weight. Common translation acts as $Ja = (a, a)$. The cost is

$$(dm + dd + a)^2 + (dm - dd + a)^2 = 2(dm + a)^2 + 2(dd)^2. \tag{3.19}$$

It is minimized at $A^F = -dm$, leaving

$$h^{\min} = 2(dd)^2. \tag{3.20}$$

Common location change is removed while change in their relative separation remains. This calculation is a realization of the general construction; the construction itself applies to any admitted smooth isometric group action.

If some nonzero ξ moves none of the laws, then $J\xi = 0$ and the normal equation cannot determine that connection component. At constant orbit rank it determines a smooth affine class modulo this infinitesimal stabilizer. When the infinitesimal stabilizer is zero the solution is unique and obeys the affine patch law. A free proper action additionally gives the usual smooth statistical orbit space. These conclusions concern different hypotheses [21]. The stabilizer typically jumps where laws agree. For one unit-covariance normal law in the plane with mean (ϵ, c) under rotations, the selected connection is $-\epsilon/(c^2 + \epsilon^2)dc$; its integral is $-\pi$ for every $\epsilon > 0$, so it concentrates at the rotation-invariant point as $\epsilon \to 0$. The minimized tensor stays bounded there, while the connection and its curvature do not.

Orthogonality also measures the cost of choosing another connection:

$$h^A = h^{\min} + \mathfrak{k}(A - A^F, A - A^F), \qquad \mathfrak{k}(\xi, \eta) = \mathsf{G}(J\xi, J\eta). \tag{3.21}$$

The minimized tensor is positive semidefinite, and its rank is at most the number of shape directions. For a single law whose group acts transitively, such as one Gaussian under the affine group, there are none; a single agent's contextual tensor then records only the supplied connection. Whether a collective tensor has enough independent directions, and whether inference preserves those directions, depends on the actual section and its dynamics. The least-squares rule alone supplies neither persistence nor a physical interpretation of the contextual base.

3.7 What compression does to this geometry

A fixed normalized channel K transforms a family P_θ into $P_\theta K$. The output score is the input score averaged over possibilities compatible with the output. The law of total covariance gives

$$I(P_\theta) - I(P_\theta K) = \mathbb{E}_\theta \operatorname{Cov}_\theta(\ell_\theta(X) \mid Y). \tag{3.22}$$

The missing Fisher information is exactly the uncertainty remaining about the score. A zero value recovers that local score, not necessarily the full family of laws [25]. If the channel itself varies with θ, its own score must be added before conditioning. The output can then acquire information through the channel parameter, so contraction relative to the input alone need not hold.

Even a fixed equivariant channel need not preserve the collectively selected connection. In the two-location example, take $(\mu_1, \mu_2) = (x, -x)$. The fine metric is $\operatorname{diag}(1, 1)$ and its collective connection is zero. Add independent unit-variance normal noise to the first observation. The first variance becomes two, so the output Fisher metric is $\operatorname{diag}(1/2, 1)$. Its connection minimizes

$$\frac{1}{2}(1 + a)^2 + (-1 + a)^2, \tag{3.23}$$

whose derivative is $3a - 1$. The new minimum is $a = 1/3$. The channel has reduced distinguishability and commutes with translation, yet the weighting of shape change against translation has changed.

In general, let a smooth equivariant retention R map the fine family to the coarse family, with metrics g, g_c. Set $\Delta = g - R^* g_c$, and let r be the fine horizontal residual. Because $dRJ = J^c$, the fine connection also minimizes the coarse cost exactly when

$$\Delta(r(X), J\xi) = 0 \quad \text{for every } X, \xi. \tag{3.24}$$

This follows by inserting the fine connection in the coarse normal equation and using fine orthogonality. It is a condition on the mixed shape–orbit part of information loss. Preserving all orbit scores is sufficient for a fixed channel, but contraction alone is not. The minimized tensor itself always contracts under a fixed channel, with an exact account of the loss:

$$h^{\min} - h_c^{\min} = \Delta(r, r) + \mathfrak{k}_c(A^F - A_c^F, A^F - A_c^F). \tag{3.25}$$

The first term is information discarded along the shape residual; the second is the shift of the selected connection, weighted by the coarse locked inertia. In the example above, $2 = \frac{4}{3} + \frac{1}{2} + \frac{1}{6}$.

For independently chosen fine and coarse connections there is another contribution. A bundle map may send a fine horizontal vector to a vector that is not coarse-horizontal. If a denotes that vertical difference and $u = D^\omega \zeta$, the coarse derivative is $T^V \Psi u + a$. Expanding its squared norm gives the fine information loss minus two cross terms and minus $\|a\|^2$. Zero horizontal mismatch is sufficient for contraction of the base tensor; a nonzero mismatch can still be dominated by information loss. The actual derivative and both connections must therefore be specified.

3.8 Geometric structure and physical interpretation

The pullback construction is already a mathematical geometry of contextual statistical change. Its physical interpretation requires a model of the contexts, a mechanism selecting the comparison data, and observable responses to transport. Inherited evolution and inherited responses impose further conditions on a parent; connection preservation is an additional requirement.

Fisher geometry is positive semidefinite. An indefinite metric can be introduced through a separate construction. For example, a Hermitian 2×2 matrix $H = tI + \boldsymbol{x} \cdot \boldsymbol{\sigma}$ has determinant $t^2 - |\boldsymbol{x}|^2$, where $\boldsymbol{\sigma}$ are the Pauli matrices. Pulling its determinant bilinear form back by an invertible soldering field, and dividing by a positive scale that transforms as $|\det g|^2$, gives a Lorentzian metric invariant under the full complex frame group. The natural scale is the determinant of the aggregate second moment; the metric is then one half of a complex-Gaussian Fisher metric minus the square of the entropy rate, so time-like directions are those in which the entropy changes faster than the law's Fisher speed. Without a scale the form is invariant only under frames with $|\det g| = 1$. A circle extension then requires a circle connection and a positive radius. Those choices do not follow from Fisher positivity, and their existence does not establish spacetime dynamics.

Chapter 4

Free Energies, Coupled Relaxation, and Source Selection

The dynamical variables are probability laws. Moving an agent therefore means changing its expectations, uncertainties, and dependencies, rather than moving a point through its sample space. The physical analogue developed here is overdamped motion on a probability manifold: an objective supplies the driving covector, and the Fisher metric converts that covector into a velocity. Neither ingredient determines the other.

4.1 Three objectives

Before any equation, one must say what is being minimized, and for interacting agents there are three different answers. In the global picture all agents are parts of one normalized joint law over their contents and records, as the spins of one lattice share one Boltzmann weight. Each agent's own free energy is then that single functional seen as a function of its own factor, peers act on it only through expectations of their contents, and every local update lowers the same total: coordinate descent, like a sequence of single-spin heat-bath moves.

In the per-agent picture each agent has its own generative model, in which its own evaluator and the laws its peers communicate compete as sources of prior. The peers' laws are data for the agent, like an external field it does not control. Each agent lowers its own free energy, but the free energies of the agents that listen to it change through terms it does not control. Such non-reciprocal forces need have no global Lyapunov function: on a directed ring of three Bernoulli agents, each pulled toward its successor, the linearized motion spirals, with complex eigenvalues that no gradient flow can have.

The third object, developed in this chapter, is the reciprocal law potential. It adds the divergences between agent laws directly, so each edge pulls its receiver toward the source and also pulls the source toward its receivers, like a spring that acts on both ends. All agents then descend one shared function, and the dynamics dissipates that function and cannot spiral into an equilibrium. The price is that this function is not the free energy of any fixed normalized model in which the agent laws are the beliefs, and records enter it only as separate assimilation events. The constructions of the later chapters use this third object; their conclusions are not claims about the other two.

4.2 The potential and its source rows

At a fixed context, let q_i describe agent i's represented state k, and let s_i describe its model presentation m. For this construction their joint recognition law is the explicit product $q_i \otimes s_i$.

Choose a model reference $r_i(dm)$ and a normalized evaluator $G_i(m, dk)$. With the reference joint $R_i(dk, dm) = r_i(dm)G_i(m, dk)$, their local contribution is

$$\mathcal{E}_0 = \sum_i D_{\mathrm{KL}}(q_i \otimes s_i \| R_i) = \sum_i \left[D_{\mathrm{KL}}(s_i \| r_i) + \int D_{\mathrm{KL}}(q_i \| G_i(m, \cdot)) s_i(dm) \right]. \tag{4.1}$$

The equality separates the model marginal from the conditional state law. Thus a product recognition family still has coupled dynamics: changing s_i changes the evaluator against which q_i relaxes, and changing q_i changes the model costs.

The channels of chapter 2 send peer laws into comparable spaces. Write $\nu^q_{ij} = q_j \mathcal{K}^q_{ij}[\mathbf{q}]$ and $D^q_{ij} = D_{\mathrm{KL}}(q_i \| \nu^q_{ij})$, where $\mathbf{q}$ collects the belief laws. Define ν^s_{ij} and D^s_{ij} similarly for model laws. With normalized source rows β_i, γ_i, positive reference rows π^q_i, π^s_i, and $T_q, T_s > 0$, the potential is

$$\begin{aligned} \mathcal{E} = \mathcal{E}_0 &+ \sum_i \left[\sum_j \beta_{ij} D^q_{ij} + T_q D_{\mathrm{KL}}(\beta_i \| \pi^q_i) \right] \\ &+ \sum_i \left[\sum_j \gamma_{ij} D^s_{ij} + T_s D_{\mathrm{KL}}(\gamma_i \| \pi^s_i) \right]. \end{aligned} \tag{4.2}$$

Each sum uses its declared source set. The row penalty balances preference for compatible peers against departure from the reference routing.

Hold the laws fixed. For either sector, minimize $f(\eta) = \sum_j \eta_j D_j + T D_{\mathrm{KL}}(\eta \| \pi)$ subject to $\sum_j \eta_j = 1$. Stationarity gives $D_j + T[\log(\eta_j/\pi_j) + 1] = \lambda$, hence

$$\eta^*_j = \frac{\pi_j e^{-D_j/T}}{Z}, \qquad Z = \sum_j \pi_j e^{-D_j/T}, \qquad f(\eta) = -T \log Z + T D_{\mathrm{KL}}(\eta \| \eta^*). \tag{4.3}$$

The last identity proves that this stationary row is the unique minimum. Eliminating both sets of rows gives $\mathcal{E}_{\mathrm{red}} = \mathcal{E}_0 - T_q \sum_i \log Z^q_i - T_s \sum_i \log Z^s_i$. At fixed temperatures and reference rows,

$$d\mathcal{E}_{\mathrm{red}} = d\mathcal{E}_0 + \sum_{ij} \beta^*_{ij} dD^q_{ij} + \sum_{ij} \gamma^*_{ij} dD^s_{ij}. \tag{4.4}$$

This follows by differentiating each log partition sum. Optimized row derivatives disappear; derivatives inside their costs remain.

4.3 Every interaction role contributes to motion

For normalized densities a, b, variation of $\int a \log(a/b)$ gives

$$dD_{\mathrm{KL}}(a \| b) = \int \log(a/b) da - \int (a/b) db, \qquad d\nu^q_{ij} = (dq_j)\mathcal{K}^q_{ij} + q_j D\mathcal{K}^q_{ij}(d\mathbf{q}). \tag{4.5}$$

Here da, db denote mass-preserving variations, and differentiation assumes a regular dominated family. An agent moves through its receiver role, its source role, and any dependence of the

channels on its law. Directed routing does not remove incoming contributions to source motion. The source is pulled toward the pooled law of the agents that listen to it, so a source more confident than its listeners' spread is widened; a communicating population realizes this only if listeners send a reverse message back to their sources. For an identity self-channel, the two laws remain identical and their divergence stays zero. Holding an old self snapshot fixed instead creates a different potential after the current law changes.

For positive finite probability tables, collect all dynamical tables as u^a, including the source rows when they are not eliminated. The Fisher metric and its unit-mobility descent are

$$\begin{aligned} g_u(v,v) &= \sum_{a,z} \frac{(v_z^a)^2}{u_z^a}, \\ \dot{u}_z^a &= -u_z^a(h_z^a - \mathbb{E}_{u^a} h^a), \qquad h_z^a = \frac{\partial \mathcal{E}}{\partial u_z^a}, \\ \dot{\mathcal{E}} &= -\sum_a \mathrm{Var}_{u^a}(h^a). \end{aligned} \tag{4.6}$$

Centering preserves each normalization. Substituting the velocity into $\dot{\mathcal{E}} = \sum_{a,z} h_z^a \dot{u}_z^a$ proves the dissipation identity. The same construction uses the Fisher tensor on a regular smooth family [1]. Both sectors have equal mobility; a slower model response must arise from the resulting dynamics. Equal mobility is what one gets by treating all tables as one product law of all represented variables and source labels, since Fisher information adds over independent factors. It has a cost for the source rows: under unit mobility a row's log odds relax at a rate equal to its temperature, so a row that selects sharply moves slowly, and replacing it by its instantaneous Gibbs optimum is a fast-row limit rather than the same flow.

A Bernoulli law makes the metric concrete. Its success probability p has metric $dp^2/[p(1-p)]$, so the mobility is $p(1-p)$. For $\mathcal{E}(p) = D_{\mathrm{KL}}(\mathrm{Ber}(p) \| \mathrm{Ber}(p_0))$, with $p, p_0 \in (0,1)$, write $l = \log[p/(1-p)]$ and $l_0 = \log[p_0/(1-p_0)]$. Then

$$\dot{p} = -p(1-p)(l - l_0), \qquad \dot{l} = -(l - l_0). \tag{4.7}$$

Log odds relax exponentially, although probability does not follow a linear restoring equation. Here t parametrizes the chosen relaxation, and the dimensionless T_q, T_s weight routing penalties. They have not been identified with physical time or bath temperature.

4.4 A record likelihood defines a different objective

A fixed normalized prior P_0 and record likelihood ℓ_o give posterior $\Pi_o = \ell_o P_0 / Z_o$ and the exact identity

$$\mathcal{F}(Q; o) = D_{\mathrm{KL}}(Q \| P_0) - \mathbb{E}_Q \log \ell_o = D_{\mathrm{KL}}(Q \| \Pi_o) - \log Z_o. \tag{4.8}$$

Source inference in a product recognition family then uses expected negative log likelihood as its cost. Peer coupling instead uses divergence between laws, whose source template can move during relaxation. Both yield Gibbs rows because both contain a categorical relative-entropy penalty, with coefficients T_q and 1 respectively; their costs and state derivatives differ. There is one setting

in which the two meet exactly. If each agent receives n messages from a source chosen by its prior and registers whether their empirical frequencies match its own law, the posterior over which source was heard is the Gibbs row at temperature $1/n$. The row temperature is then the reciprocal number of exchanged messages, as in a counting experiment in which a sharper verdict needs more counts. Identical independent Gaussian laws have zero peer divergence but nonzero expected squared sample difference. A normalized likelihood for that difference therefore describes an interaction distinct from direct agreement of the laws.

Chapter 5

What a Message Lets the Receiver Infer

Sending a few observables can change an agent's description of many variables. The extra structure comes from the receiver's conditional model. This is familiar in physics: measuring one coordinate of a correlated system can change predictions for another coordinate without measuring it directly. The distinction between transmitted information and inferred structure is explicit here.

5.1 Selective sending and conditional reconstruction

A sender uses a measurable map $T_j : E_j \to \mathsf{H}$ into a message space. If its content has law q_j, possible messages have law

$$\mu_j = (T_j)_{\#} q_j. \tag{5.1}$$

The pushforward symbol means $\mu_j(A) = q_j(T_j^{-1}A)$. Sampling a message and transmitting the whole law μ_j are different communication protocols. Messages run from sender to receiver. The reciprocal potential of the dynamics chapter also moves each sender toward its listeners, which a communicating population can only realize if the listeners send something back. A linear receiving map $O_i : \mathsf{H} \to E_i$ embeds a message, but its composition with T_j cannot have higher rank than T_j.

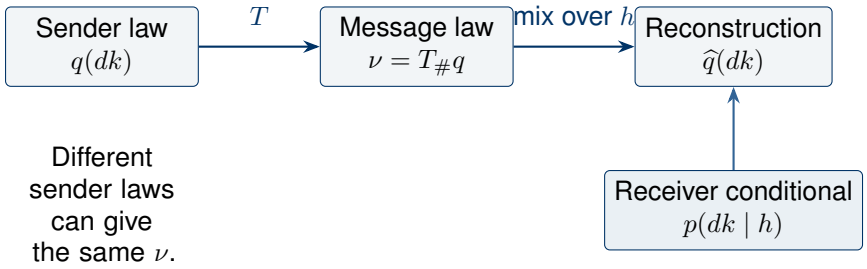


Figure 5.1: Selective communication and receiver structure. The sender fixes the transmitted marginal. The receiver supplies a conditional law for distinctions that the message does not specify. This distributional reconstruction presupposes that the message law is available; one received sample supplies a value instead.

To see exactly what is lost, work in one common presentation and write $T : E \to \mathsf{H}$. Let q be the sender's law and p the receiver's reference law, with finite $D_{\mathrm{KL}}(q\|p)$. The receiver can combine the transmitted marginal $\mu = T_{\#}q$ with its own conditional law:

$$\widehat{q}(dk) = \int p(dk \mid h)\mu(dh). \tag{5.2}$$

Disintegration on standard Borel spaces provides these conditional laws [17]. The reconstruction has exactly the right message marginal. Its remaining discrepancy is

$$D_{\mathrm{KL}}(q\|\widehat{q}) = \int D_{\mathrm{KL}}\big(q(\cdot \mid h)\|p(\cdot \mid h)\big)\mu(dh). \tag{5.3}$$

Both laws assign the same probability to h, so that factor cancels in their density ratio. The remaining logarithm compares the two conditional laws, which gives the equality by integration. This reconstruction is Jeffrey's rule of conditioning [10], and it is the law closest to the receiver's prior among all laws with the transmitted marginal. The divergence then splits like a right triangle, $D_{\mathrm{KL}}(q\|p) = D_{\mathrm{KL}}(q\|\widehat{q}) + D_{\mathrm{KL}}(\widehat{q}\|p)$, with the second leg equal to the divergence of the message marginals [8].

For a calculation on four points, let $k = (H, V)$ be two bits. The sender gives probability $1/2$ to each of 00 and 11, and transmits only H. A receiver whose reference makes both bits independent and fair reconstructs the uniform law on all four points. The message marginal is correct, while the relative entropy is $\log 2$. A receiver whose conditional model has $V = H$ reconstructs the sender's full law. The missing correlation was supplied by the receiver.

One reference conditional can recover a whole family only under a stronger condition. Suppose $q_\alpha = L_\alpha p$ and use the same $p(dk \mid h)$ for every α. Its reconstruction has density

$$\frac{d\widehat{q}_\alpha}{dp}(k) = \mathbb{E}_p[L_\alpha \mid T](k). \tag{5.4}$$

To obtain this identity, weight the reference message marginal by its conditional mean likelihood and then apply the reference conditional. The result equals q_α precisely when L_α is already measurable from the message, which is the classical statement that the message is a sufficient statistic for the family [15]. Reconstructing one posterior with its own conditional does not establish this family-wide property. If a local reconstruction must also ignore retained exterior information, the reference needs conditional independence from that exterior given the message.

For a single received value h, Bayesian assimilation instead takes the form

$$q_i^+(dk) = \frac{\ell_i(h \mid k)q_i^-(dk)}{\int \ell_i(h \mid k')q_i^-(dk')}, \tag{5.5}$$

with a positive finite denominator. In the perfectly correlated two-bit example, a noiseless observation $H = 1$ also fixes $V = 1$. This is inference through an existing relation, not an inversion of an arbitrary lossy map. Under the reciprocal potential such an update is a kick: the flow afterwards relaxes toward states that do not depend on the record, unless the record's likelihood is kept in the objective.

5.2 Agreement needs a common prediction rule

Suppose a model law s and a normalized record evaluator $K(m, do)$ predict the record law sK. If two agents compare model laws in one frame, they must also compare their evaluators in that frame. Equal model laws can predict opposite bits when their evaluators differ.

For one common evaluator and finite $D_{\mathrm{KL}}(s\|t)$, an exact information balance is

$$D_{\mathrm{KL}}(s\|t) = D_{\mathrm{KL}}(sK\|tK) + \int D_{\mathrm{KL}}(s^o\|t^o)(sK)(do), \tag{5.6}$$

where s^o, t^o are the conditional model laws after record o. The joint laws $s(dm)K(m, do)$ and $t(dm)K(m, do)$ have divergence $D_{\mathrm{KL}}(s\|t)$ because their common kernel cancels. Factoring the same joint laws by their record marginal gives the displayed split. The last term is the distinction left unresolved by the record. A constant evaluator discards every distinction; a sufficient record leaves none.

For a model link Θ^s_{ij}, compatibility means $K_i(\Theta^s_{ij}m, A) = K_j(m, A)$ after identifying record alphabets. Only then does transporting the model law also transport its predictive meaning. If the link changes with the laws, this condition must continue to hold at the updated state.

5.3 Compression should name what matters

Rate–distortion theory makes the desired fidelity part of the communication problem [27]. A low-dimensional real vector alone is not a finite-bit channel. Noise, precision, or an encoding constraint must specify the communication cost. Wyner–Ziv coding additionally gives the receiver a realized variable correlated with the source [33]; possession of a prior by itself is not that realized side information.

For a predictive criterion, name the source X, target response Y, and accessible receiver information Z. An encoder generates U from X alone, so $U - X - (Y, Z)$ is a conditional-independence chain. The optimal decoder is $P(Y \mid U, Z)$. Its expected relative-entropy distortion is

$$\mathcal{D} = I(Y; X \mid U, Z) = I(Y; X \mid Z) - I(Y; U \mid Z). \tag{5.7}$$

The equality follows by expanding $I(Y; X, U \mid Z)$ in both orders. It is the information about the chosen response that compression discards. An information-bottleneck criterion trades this quantity against a cost such as $I(X; U \mid Z)$ [30]; an operational bit rate requires additional coding assumptions.

If $X = (A, B)$ consists of independent fair bits and $U = A$, the distortion is zero for target $Y = A$ and $\log 2$ for target $Y = B$. For a prospective parent, the target can be a future boundary response. Retaining one such response does not ensure that every later interaction can be inherited. That stronger requirement concerns dynamics.

Chapter 6

When a Collective Can Act as an Agent

A collective can have a simple description without being able to evolve from that description alone. Knowing the total energy of two bodies, for example, does not usually determine their next interaction. A parent of law-valued agents faces the same question: does its retained state determine its future responses?

There are two operations to distinguish. A channel C compresses represented contents and gives a parent law $Q_x C$. A map R retains a complete dynamical state $R(x)$. Besides belief and model laws, that state can include the model evaluator, source information, boundary channels, and memory. A distribution over possible contents does not automatically contain these additional data.

6.1 Two routes to the same parent

Suppose the fine configuration advances by F. Exact inherited evolution requires a parent update F^c such that

$$R(F(x)) = F^c(R(x)). \tag{6.1}$$

We can first evolve the children and then form their parent, or first form the parent and evolve it. Both routes must give the same retained state. Repeating the identity gives agreement at every later step. Boundary response mechanisms must likewise depend only on the retained state.

The immediate test is whether configurations with the same parent have the same next parent. If $R(x) = R(x')$ but $R(F(x)) \neq R(F(x'))$, no single F^c can satisfy the equation. Conversely, in a finite state space, this test is sufficient: evolve any representative of each parent state. Agreement within each fiber of R makes the choice irrelevant. On general measurable spaces the resulting map must also be measurable.

For stochastic dynamics, replace the next state by its distribution. If T is the fine transition matrix, the probability

$$\sum_{y:R(y)=z'} T(x, y) \tag{6.2}$$

must depend only on $R(x)$. These sums then define normalized parent rows. Conditioning on any parent history merely averages identical rows, so the parent reproduces the whole retained history law. This is strong lumpability [18]. Agreement for one equilibrium preparation is a weaker statement.

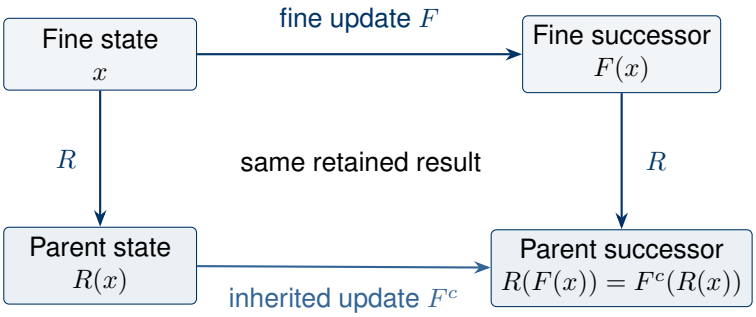


Figure 6.1: The two routes to the next parent configuration must agree. Vertical maps retain information; horizontal maps advance the dynamics. A compressed law alone need not make this square commute.

6.2 When a gradient flow descends

Suppose the fine configuration lies on a Riemannian manifold with metric g and evolves by $-\operatorname{grad}_g E$. Let a smooth surjective submersion R retain the parent, whose metric is g_c. One sufficient condition is that the energy depend only on the parent,

$$E = E_c \circ R, \tag{6.3}$$

and that R preserve lengths of horizontal vectors, meaning vectors orthogonal to $\ker DR$. Then the gradient is horizontal because the energy has no derivative along a fiber of R. For a horizontal u,

$$g_c(DR \operatorname{grad}_g E, DRu) = g(\operatorname{grad}_g E, u) = dE_c(DRu). \tag{6.4}$$

Every parent tangent has a horizontal lift, so the pushed gradient is exactly $\operatorname{grad}_{g_c} E_c$. The dynamics descends in the same time.

If instead $E = \chi(E_c \circ R)$ and the horizontal metric is scaled by $d(x)^2$, the pushed vector field acquires the factor $\chi'(E_c(Rx))d(x)^2$. A positive factor preserves oriented trajectories after a time change. A factor that depends on discarded state does not specify an autonomous parent clock. Taking the minimum of E over each fiber does not supply the first hypothesis: it gives the value at the minimizing states, not the fine energy everywhere on that fiber.

The requirement $E = E_c \circ R$ is stronger than descent needs, and it has a physical cost: it makes the energy blind to every motion along a fiber, so the discarded coordinates cannot move relative to the retained ones. Only the horizontal derivatives matter. The computation above uses $dE(u)$ for horizontal u alone, so descent holds exactly when

$$dE(u) = dE_c(DR(u)) \qquad \text{for every horizontal } u. \tag{6.5}$$

Two particles on a line with unit masses, harmonic self-energies $\frac{1}{2}x_1^2 + \frac{1}{2}x_2^2$, and any pair interaction $W(x_1 - x_2)$ show the difference. The center of mass $(x_1 + x_2)/\sqrt{2}$ relaxes as if the interaction were absent, because a force that depends only on the separation pushes the two particles in opposite directions and cancels in their sum. Meanwhile the separation evolves under W. The parent is exact, the discarded coordinate has its own dynamics, and yet the energy is not a function of the center of mass.

The opposite case is a retention that simply drops some agents. If the retained and discarded agents exchange any peer cost, the retained velocity depends on the discarded state, because each agent pulls on every agent it attends to and is pulled back. Dropping whole agents is exact only for blocks that do not interact at all. Exact compression of interacting agents therefore comes from symmetry, as when every member of a block can be rotated together without changing any cost, or from special energy structure such as the center-of-mass cancellation. A related consequence is that exact inheritance cannot create oscillation: the linearization of a gradient flow at an equilibrium has real eigenvalues, and an exact parent inherits a subset of them.

6.3 A missing role can prevent closure

Let two independent represented bits have laws $\mathrm{Ber}(p_1)$ and $\mathrm{Ber}(p_2)$. Their probabilities evolve according to

$$\begin{aligned} \dot{p}_1 &= -p_1 + \lambda(p_2 - p_1), \\ \dot{p}_2 &= -2p_2 + \lambda(p_1 - p_2), \qquad \lambda \geq 0. \end{aligned} \tag{6.6}$$

The square $[0,1]^2$ is preserved: the vector field never points outward on its boundary. It is tangent to a face when $\lambda = 0$ and vanishes at the corner $(0,0)$. Retain only the law of the sum of the two bits. The configurations $(0.8, 0.2)$ and $(0.2, 0.8)$ both give probabilities $(0.16, 0.68, 0.16)$ for sums $0, 1, 2$.

Their subsequent changes differ. The coupling terms cancel when we add the two equations, giving

$$\frac{d}{dt}\mathbb{E}(K_1 + K_2) = -(p_1 + 2p_2). \tag{6.7}$$

The derivatives are -1.2 and -1.8. Thus the same retained law has two different possible derivatives. Compression erased which probability belonged to the faster-decaying role. The fine recognition law remains a product throughout. What must be retained is the information this evolution uses; cross-agent covariance is not automatically required.

6.4 An exact supplied hierarchy

A finite construction shows that two successive reductions can work. Take $x = (a, b, c, r) \in \{0,1\}^4$, with internal register r and readout $\rho(x) = r$. Let the agent laws be

$$q_a = \mathrm{Ber}\left(\frac{1+2a}{4}\right), \qquad s_a = \mathrm{Ber}\left(\frac{2+a}{4}\right). \tag{6.8}$$

The model evaluator assigns probability $1/4 + m/2$ to record value one when the model presentation is $m \in \{0,1\}$. Averaging over s_a gives the predictive law $\nu_a = \mathrm{Ber}(1/2 + a/8)$.

At each step draw independent bits ξ, η, υ with probabilities $1/4, 1/3, 1/5$ of being one, independently of previous draws, and set

$$\begin{aligned} a' &= a \oplus \xi, \\ b' &= b \oplus a \oplus \eta, \\ c' &= c \oplus b \oplus \upsilon. \end{aligned} \tag{6.9}$$

Here $\oplus$ means addition modulo two. Independently of these draws, conditional on the current state, give r' law ν_a. The record is therefore an internal state change. Its generation rule is supplied, rather than deduced from the existence of an evaluator.

First retain (a, b, r), then retain (a, r). Summing the fine transition over c' removes a normalized factor and leaves a row independent of c. Summing that row over b' removes another normalized factor and leaves a row independent of b. The configuration spaces shrink from sixteen to eight to four states. Both parents retain q_a, s_a, their evaluator, and the register. They reproduce the complete retained state and record histories for every initial law. The coupling is real, since a affects b' and b affects c'; its triangular form permits these reductions. This is a drive and its response: a drives b, which drives c, and nothing acts back. Each retained block is autonomous because the variables it keeps never read the variables it discards. The configuration describes one agent with internal marks, so neither level groups two agents, and the hierarchy itself was prescribed.

6.5 What a parent must remember

When closure fails, a parent can retain more state or, under a declared stochastic process, the conditional law of hidden states given its accessible history. In a finite model this is a probability vector: propagate it with the fine transition, restrict to the new retained value, and normalize. That vector determines every future response even when the current summary does not. Keeping only an equilibrium hidden distribution generally changes transient predictions. The vector lives on the simplex of all fine states, so it saves nothing unless noise or poor observability keeps the history from pinning down the fine state.

There is a middle case in which the parent needs no memory although the current summary is not autonomous for every preparation. Suppose that for each retained value z there is a fixed way $\Lambda(z, \cdot)$ of spreading probability over the fine states with that value, and that spreading and then evolving gives the same result as evolving the parent and then spreading, $\Lambda T = T^c \Lambda$. A child population prepared by spreading a parent law in this way stays spread in this way forever: the hidden distribution is always $\Lambda(Z_n, \cdot)$, and the parent is an ordinary Markov chain with kernel T^c [26]. A handoff that prepares children differently can break the parent, even though the kernels are unchanged.

Given the fine update and a list of what the parent must reproduce (its laws, the responses to a declared set of probes, and the channels it reads), there is a smallest exact parent. Two configurations are lumped together exactly when the listed quantities agree for them now and at every later time. Nothing coarser can be exact, and anything exact remembers at least this much. A longer list of probes gives a finer parent; if outside agents may probe the children without restriction, they can tell every exposed child law apart and the parent cannot forget any of them. How much a parent forgets is therefore decided by the probes declared before the parent is chosen.

There is an exact way to separate insufficient memory from inaccurate prediction. Let P be a declared law of a finite record sequence $Z_0, \ldots, Z_{T-1}$ on standard Borel spaces. A retained statistic is $M_t = f_t(Z_{<t})$. Let S use normalized prediction rows $s_t(\cdot \mid M_t)$. The path-law chain rule and

conditional disintegration give

$$D_{\mathrm{KL}}(P\|S) = \sum_{t=0}^{T-1} I_P(Z_t; Z_{<t} \mid M_t) + \sum_{t=0}^{T-1} \mathbb{E}_P D_{\mathrm{KL}}\big(P(Z_t \in \cdot \mid M_t)\|s_t(\cdot \mid M_t)\big). \tag{6.10}$$

The first sum is information about the next record that was discarded with the history. The second is error in the chosen prediction given the retained information. Both are nonnegative, including when the total is infinite. This extends the finite-window Markov decomposition [13, Thm. 7.3.1]. Zero first sum gives predictive sufficiency for the declared process; a compact recursively updated memory still needs a realization $M_{t+1} = U(M_t, Z_t)$. Changing the allowed controls changes which histories must be predicted.

Statistical dependence alone also does not select an agent hierarchy. A declared finite cross-agent joint law has a finest partition into independent blocks, but arbitrarily weak couplings can merge those blocks. A deterministic selector reading only a fully permutation-symmetric law cannot choose a nontrivial intermediate hard partition while respecting relabeling: swapping one member with an outside agent would create a different block that overlaps the original. More generally, such a selector can only output a partition that every symmetry of the law preserves; on a ring of four agents with the symmetries of a square, it can pair opposite agents but never neighbors. Realized histories, additional geometry, or a distribution over partitions can fall outside that restriction. The result identifies what a selector must read; it does not identify independence with agency.

6.6 The formation standard

The theory aims to show that interacting agents can produce further agents. A construction that groups agents and hands their laws to parents does not yet show this, because the grouping and the handoff can be arranged without the interaction doing anything. The formation standard of the main text lists six clauses, each closing one such loophole.

The first clause asks for a set of starting configurations that is open, or a family declared in advance and preserved by the dynamics. A single trajectory is not enough. Along one path, almost any summary of the state traces out a curve, and every curve is the solution of some differential equation, so "the parent evolves autonomously" would hold for nearly any choice of parent. Exactness means something only when it is tested on configurations that the parent cannot tell apart. The second clause asks the dynamics to pick nontrivial groups that persist by a declared criterion, with the moment of handoff read off from the state or its accessible history. The third asks that each group carry inherited laws, evaluators, responses, and dynamics, with the probes that outsiders may use fixed before the parent is chosen.

The fourth clause is compression. A parent that simply keeps every child's state is always exact and says nothing. Four agents with Gaussian laws pulled together by symmetric peer costs make the point: a threshold rule first pairs the two closest agents, then the other two, then the two pairs, and keeping the full tuple of child states at each stage reproduces the dynamics exactly. Nothing

has been forgotten, so nothing has formed. The clause therefore asks each parent to lump together configurations that some admitted probe of the children could still distinguish.

The fifth clause is the control experiment. Delete every cost that couples one agent to another, freeze the population dependence of the channels at its initial value, keep everything else, and run the same selector. This is the null flow. If it produces the same groups, the interaction did no selection work. Agents that do not interact can be grouped by any rule that reads their initial data and a common relaxation clock, much as runners in a race can be grouped by finishing order even though they never touch. The standard asks that the sequence of selected groups differ from the null one on an open set of starting configurations. Only the groups count: if the interaction merely makes the same joins happen sooner or later, it has changed a clock, not the outcome. A stronger version asks that the final grouping not be predictable from any summary of the initial data that evolves the same way with and without interaction, and an amplifying version asks that groups be tighter relative to their separations than they would be without interaction.

A whole class of architectures fails this control. If every receiver compares itself with the same target law, and the channels see the population only through additive moments, then each pair of agents relaxes toward each other at a common rate: the logarithm of the ratio of their laws shrinks by the factor e^{-2t}, up to a constant. Without the peer costs the same quantity shrinks by e^{-t}. The interaction then only speeds up a clock that every pair shares, and any selector that compares pairs in this way produces the same sequence of groups with or without it, at different times. The same happens whenever summing two agents' parameters is an exact parent for every pair: the parameter equation must then be affine, all differences shrink by a common factor, and the order of joins is fixed by the initial distances. Selection that depends on the interaction needs receivers that weigh different sources differently.

The sixth clause is recursion. The parents must themselves be grouped, by the inherited interactions, without resetting their data or adding a new attraction tuned for the purpose, and this second grouping must depend on what the first stage produced. If it would occur from every possible handoff in the inherited class, the second stage is a sequence of scales supplied in the initial data, not a parent built from parents.

A construction that meets the selection, inheritance, and recursion clauses but fails compression or the null-flow control is called a two-stage selection with exact inheritance. The constructions in the later chapters are measured against this standard. The nonsmooth fusion example meets the null-flow control, since without coupling its agents never collide, at the cost of changing the interaction. The common-precision matrix-law construction of chapter 11 selects two parents and then their union, but its schedule does not depend on the coupling: it is a two-stage selection with exact inheritance. The six-agent construction of chapter 12 works on a full open neighborhood of laws with selective channels, but in the regime of its proof the null flow reproduces its membership history. Both smooth constructions supply their compression architectures in advance. A construction that meets every clause, with a selector and a compression interface derived rather than declared, remains open.

Chapter 7

A Nonsmooth Route to Finite Coalescence

Smooth attraction can bring two laws progressively closer without making them equal at any finite time. An absolute-difference interaction behaves differently: its force does not vanish as the gap closes. This chapter explains a nonsmooth modification of the reciprocal law potential in which agents coincide at finite times and their coincident blocks inherit laws, evaluator, metric, and responses exactly. Four agents can fuse into two parents that subsequently fuse under the inherited interaction. The grouping follows the evolving state; the equations contain no prescribed pairs. The outcome is always a finite coalescence tree that ends in one block.

Pairwise convex fusion penalties and finite-time nonsmooth consensus have established precedents [7, 16], and on the restricted family used here each sector is a weighted particle solution of the one-dimensional aggregation equation with kernel $\lambda|x|$, whose sticky collisions and finite-time collapse are known [3, 5]. The additional task here is to inherit the probability laws, model evaluator, metric, and subsequent responses when fusion occurs.

7.1 Laws, a model, and inherited multiplicity

Write $\mathrm{LN}(\mu, \sigma^2)$ for the law of a positive random variable whose logarithm is normal with mean μ and variance σ^2. A component containing n primitive agents carries

$$q_n = \mathrm{LN}(\mu^q, 1/n), \qquad s_n = \mathrm{LN}(\mu^s, 1/n), \qquad n \geq 1. \tag{7.1}$$

The variances are held fixed; this is a constraint, not a family that a variance-inclusive flow would preserve, and releasing the variance would destroy the encoding of n. Recognition is the product $q_n \otimes s_n$. Its normalized reference model p_n is jointly Gaussian in the logarithms, with zero mean and precision

$$n \begin{pmatrix} a & -c \\ -c & a \end{pmatrix}, \qquad a > c > 0. \tag{7.2}$$

Thus the model couples the two sectors even though recognition is factorized. Its normalized conditional evaluator is

$$G_n(m, dk) = \mathrm{LN}\left(\frac{c}{a}\log m, \frac{1}{an}\right)(dk). \tag{7.3}$$

Direct Gaussian relative entropy gives

$$D_{\mathrm{KL}}(q_n \otimes s_n \| p_n) = C + nU, \qquad U = \frac{a}{2}\left[(\mu^q)^2 + (\mu^s)^2\right] - c\mu^q\mu^s, \tag{7.4}$$

where $C = a - 1 - \frac{1}{2}\log(a^2 - c^2)$. The mixed term makes the belief update depend on the model coordinate and conversely. The Fisher metric of this recognition family is

$$ds^2 = n\big[(d\mu^q)^2 + (d\mu^s)^2\big]. \tag{7.5}$$

Both the metric and the local potential therefore carry the same multiplicity, which cancels from an agent's own velocity. The multiplicity acts as a friction weight and as an interaction charge, not as an inertia: the dynamics below is overdamped one-dimensional motion in a harmonic trap under the pair potential $n_I n_J |x_I - x_J|$.

A parent communicates at primitive resolution through the normalized channel

$$B_n(k, dw) = \mathrm{LN}(\log k, 1 - 1/n)(dw). \tag{7.6}$$

At $n = 1$ this means the identity channel. Adding the independent log variances shows that $q_n B_n = \mathrm{LN}(\mu^q, 1)$, with the analogous identity for s_n. This restores a primitive message law while retaining the narrower parent law internally. Statistically, B_n is the law of one child sample given the mean of n children, which is sufficient and does not depend on the unknown mean; that parameter independence is why fresh messages descend exactly.

7.2 The interaction that permits contact

All means here use common reference coordinates. In sector $\alpha \in \{q, s\}$, transform the source's restored message by

$$S^\alpha_{IJ}(w) = \exp\big[\chi_\alpha(\mu^\alpha_I - \mu^\alpha_J)\big]w, \qquad 0 < \chi_\alpha < 1. \tag{7.7}$$

Its log mean moves toward the receiver. Including endpoint frames gives the link $U_I \exp[\chi_\alpha(\mu^\alpha_I - \mu^\alpha_J)]U_J^{-1}$, with the usual endpoint covariance under passive re-expression. This link is pure gauge: it equals $V_I V_J^{-1}$ with $V_I = U_I e^{\chi_\alpha \mu^\alpha_I}$, so every loop is trivial, and it is a law-dependent change of frame rather than a curved transport.

The compared message variances are both one. Consequently,

$$\sqrt{2D_{\mathrm{KL}}\left(\widehat{\nu}^\alpha_I \| (S^\alpha_{IJ})_\# \widehat{\nu}^\alpha_J\right)} = (1 - \chi_\alpha)|\mu^\alpha_I - \mu^\alpha_J|, \tag{7.8}$$

where the displayed comparison is in a common reference presentation and $\widehat{\nu}$ denotes a restored law. Rescaling the coefficient by $1/(1-\chi_\alpha)$ gives an absolute-difference penalty with strength $\lambda > 0$, and χ_α then disappears from the dynamics altogether. The square root is the new constitutive choice. It changes the smooth quadratic peer penalty of the reciprocal potential into a convex function with a corner at equality.

On the complete contact graph, the energy apart from constants is

$$\mathcal{E} = \sum_I n_I U_I + \lambda \sum_{I<J} n_I n_J \left(|\mu^q_I - \mu^q_J| + |\mu^s_I - \mu^s_J|\right). \tag{7.9}$$

The coefficient $n_I n_J$ counts primitive contacts. With unit Fisher mobility the evolution is $n_I \dot{\mu}_I \in -\partial_{\mu_I} \mathcal{E}$. The subgradient of $|x|$ is its sign away from zero and the interval $[-1, 1]$ at zero; this inclusion is the same as the Filippov regularization of the almost-everywhere gradient.

Convexity makes this evolution well posed. For every initial configuration there is exactly one solution for all time, by the theory of maximal monotone operators [4]. For two solutions, the derivative of their squared distance in the fixed multiplicity metric is the negative pairing of the state difference with a subgradient difference, hence is at most $-2(a - c)$ times their squared distance. Coincident primitive agents are interchangeable in the complete-graph equations. Uniqueness then keeps them equal after contact. Sticking follows from the same dynamics that brought them together.

The corner also forces complete consensus. The largest and smallest means in a sector are pushed toward each other with a force at least λ times the total multiplicity, while the smooth part shrinks exponentially. The spread in each sector therefore reaches zero at a finite time, for every initial configuration and every set of multiplicities, and every coalescence tree ends in one block.

7.3 Two successive fusion events

Consider the invariant family

$$\mu_i^q = u_i + v, \qquad \mu_i^s = u_i - v, \qquad v(t) = v_0 e^{-(a+c)t}. \tag{7.10}$$

Both sectors have the same interagent gaps, so their interaction forces cancel in the equation for v. Set $\kappa = a - c > 0$ and introduce

$$u_i(t) = e^{-\kappa t} z_i(\tau), \qquad \tau = \frac{e^{\kappa t} - 1}{\kappa}. \tag{7.11}$$

This removes the common linear contraction. Before collisions,

$$\frac{dz_I}{d\tau} = -\lambda \sum_{J \neq I} n_J \operatorname{sign}(z_I - z_J). \tag{7.12}$$

Start four primitive agents at $z = (-3d, -2d, 2d, 3d)$, with $d > 0$. Their slopes are $(3\lambda, \lambda, -\lambda, -3\lambda)$. Each neighboring outer pair closes a gap d at relative speed 2λ. The resulting stages are

Time	Positions in z	Multiplicities
0	$-3d, -2d, 2d, 3d$	$1, 1, 1, 1$
$\tau_1 = d/(2\lambda)$	$-3d/2, 3d/2$	$2, 2$
$\tau_2 = 5d/(4\lambda)$	0	4

After the first event, the parents have slopes 2λ and -2λ. Their remaining gap $3d$ closes in another $3d/(4\lambda)$. Physical event times are $t_r = \kappa^{-1} \log(1 + \kappa\tau_r)$. No coupling or mobility is retuned at the second level. The pattern survives small perturbations off the invariant family, and with gaps separated enough in scale the same mechanism gives coalescence trees of any finite depth.

7.4 What the parent retains

At synchronization, independent log samples with a common mean have an average of variance $1/n$. Exponentiating that average gives the geometric-mean retention and the stated parent law. Restricting the primitive Fisher metric to equal location coordinates gives the factor n. Summing contact forces gives $n_I n_J$, so the reduced dynamics use exactly the inherited metric and contacts. Redundant location coordinates are removed after synchronization; an average of separated children is not claimed to close their dynamics.

Individual sample distinctions require care. If X is the retained log average, the entire primitive vector is reconstructed as

$$(X_1, \dots, X_n) = X\mathbf{1} + R, \qquad R \sim N\left(0, I - \frac{\mathbf{1}\mathbf{1}^T}{n}\right), \tag{7.13}$$

with R independent of X. The residuals sum to zero and are correlated. Independent scalar applications of B_n to one reused parent sample preserve individual marginals but give the wrong joint law.

Fresh record sampling also closes explicitly. Applying G_n to s_n gives

$$\mathrm{LN}\left(\frac{c}{a}\mu^s, \frac{(c/a)^2 + 1/a}{n}\right), \tag{7.14}$$

the law of the geometric mean of n fresh primitive model records. A primitive record instead uses a fresh restored model sample followed by G_1. These are exact statements for the declared fresh sampling protocols; they do not establish closure for sampled-record feedback into the dynamics.

7.5 How this measures against formation

Switching the interaction off is the decisive control. With $\lambda = 0$ each agent follows a linear contraction toward zero, distinct agents never meet, and no event occurs; with $\lambda > 0$ every configuration fuses. The interaction is therefore necessary for the grouping. Two other requirements fail. A fused block consists of identical copies, so retaining its common law forgets nothing that a declared probe could detect: the noninjective geometric mean acts on samples, and the forgotten sample detail is reconstructed by the parameter-free kernel above. And although the parents do select their union under the inherited interaction, that second fusion happens from every possible pair of parents, because every configuration ends in consensus; the counterfactual part of the recursion requirement fails. The construction is a two-stage selection with exact inheritance, not a recursive formation.

The comparison with the Wishart construction in chapter 11 is instructive. That construction keeps the smooth KL potential, its children remain distinct at every finite time, and its agents also approach a common limit, so the difference in eventual outcome is small. The decisive difference is the role of the interaction. There the membership schedule is the centroid dendrogram of the initial shapes and does not depend on the coupling, so switching the interaction off leaves it unchanged; here switching it off removes every event. Both are two-stage selections with exact inheritance, for

different reasons. The square-root penalty is a constitutive addition, and this example derives no geometry of the context base.

Chapter 8

Hidden Conditional Information Under Smooth Dynamics

This chapter concerns the reciprocal law potential and its Fisher flow. An exact reduction need not identify agents with one another. The requirement is that discarded distinctions have no effect on the future quantities the reduced description promises to reproduce. Under smooth KL dynamics this requirement has a concrete content: hidden conditional information can contribute to the force on a retained marginal. A per-agent reduced state must retain that contribution, or have a reason why it cancels. The first part of the chapter builds such a state completion for a single agent's private information; parents, in the sense of grouped agents, appear only in the erasure example at the end.

8.1 A hidden conditional law can move a visible marginal

Write a law as $q(dz, dv) = \mu(dz)\kappa_z(dv)$. Here z is retained and v is internal. Against a reference $\nu(dz)A(dv)$, the KL chain rule is

$$D_{\mathrm{KL}}(q\|\nu \otimes A) = D_{\mathrm{KL}}(\mu\|\nu) + \int H(z)\mu(dz), \qquad H(z) = D_{\mathrm{KL}}(\kappa_z\|A). \tag{8.1}$$

If one public value has a more costly hidden conditional law, changing its probability changes the full energy. The public Fisher velocity therefore contains $H(z)$ as well as the ordinary public force. A single average hidden entropy does not tell the reduced state which public values carry that cost.

There is an exact architecture in which this is the only hidden contribution. A sender transmits its public coordinate through a normalized, possibly law-dependent channel, and its private coordinate is not transmitted. The receiver reconstructs its private coordinate from its own conditional $\kappa_{i,z}$. The receiver law and compared source law then have the same private conditional, so that conditional cancels from their peer KL. With a fixed reference whose private factors are independent, the two-sector local model still couples the public belief and model laws. The original potential and Fisher metric give

$$\begin{aligned} \dot{\mu} &= -\mu[h^c + H - \mathbb{E}_\mu(h^c + H)], \\ \dot{\ell}_z(v) &= H(z) - \ell_z(v), \qquad \ell_z = \log(d\kappa_z/dA). \end{aligned} \tag{8.2}$$

The public score h^c includes every incoming interaction and every channel derivative. Cancellation is a functional identity, so it does not require freezing a live channel while differentiating.

The second equation contains no public law and no other agent. Each private conditional simply relaxes toward its reference along an explicit path, whatever the rest of the population does, and the public flow feels it only as a time-dependent forcing fixed at the start. In this architecture private

information never changes an interaction. That makes the construction solvable, and it is also its main limitation: the interesting case, in which private variables later influence interactions, is not covered.

8.2 Retain the distribution of density values

The conditional equation in eq. (8.2) moves a log-density value according to its current value and the conditional KL. It does not need the name of the private point carrying that value. This suggests retaining the probability law

$$\xi_z = (\ell_z)_{\#}A, \qquad \int e^r \xi_z(dr) = 1, \qquad H(z) = \int r e^r \xi_z(dr). \tag{8.3}$$

For example, use a uniform reference on two private states. The laws $(3/4, 1/4)$ and $(1/4, 3/4)$ have the same spectrum: half the reference mass sees $\log(3/2)$ and half sees $\log(1/2)$. Their private preferences differ, but their contribution to the public force agrees.

What this forgets depends entirely on the declared reference. With the reference $(0.6, 0.4)$ the same two laws have different spectra, and on a finite alphabet whose atoms have no coincidences among subset masses the spectrum identifies the conditional completely. The compression is the symmetry of the reference, not something the dynamics creates. In statistical language the spectrum is the law of the log-likelihood ratio between the conditional and the reference, so equal spectra mean that the two comparisons are equally informative in Blackwell's sense.

The complete spectrum evolves exactly. With $a = e^{-t}$,

$$\xi_z(t) = \left(r \mapsto ar - \log \int e^{ar'} \xi_z(0, dr') \right)_{\#} \xi_z(0). \tag{8.4}$$

This follows by solving $d\kappa_z(t)/dA \propto (d\kappa_z(0)/dA)^a$ and normalizing. The formula applies to arbitrary private conditionals whose log densities are bounded uniformly in the public coordinate; on general public spaces the public flow also needs bounded log densities for its reference and channels whose forces are Lipschitz in those charts. It does not fit a Gaussian, and the retained state can remain infinite-dimensional. When the initial conditionals belong to an exponential family based at the reference, however, only their natural parameters change, shrinking by the factor a, so the private part then closes on finitely many numbers.

If all public values within an agent have the same conditional spectrum, $H(z)$ is constant and disappears under Fisher centering. The public laws then follow the original coarse KL dynamics exactly. The private conditionals can still differ: a public bit can select between the two swapped binary laws above. This gives internal correlation without changing the public force. Other agents may have different public laws and different spectra; equality between agents is unnecessary. For the public flow alone even less is needed: only how the Rényi divergences of the conditionals from the reference vary with z, up to a part common to all z.

This property is not created at a finite time by the conditional relaxation. At every finite time $a > 0$, the power transformation and its normalization can be inverted. Different spectra therefore remain

different until an asymptotic limit, unless some additional operation changes them. Retaining unequal spectra is an exact alternative, but their existence alone does not select groups of agents.

The architecture matters. If the channel carried the sender's private coordinate, two agents with the same spectra could sit at different peer costs: with a uniform two-point reference, the laws $(3/4, 1/4)$ and $(1/4, 3/4)$ compared through an identity private channel cost $\frac{1}{2}\log 3$, while two copies of the first cost nothing. Separate spectra per agent then no longer determine the forces.

8.3 What a future record is allowed to inspect

The same two swapped private laws show why a boundary protocol matters. Let record 1 have probabilities $(3/4, 1/4)$ at the two private points. The record probabilities are respectively $5/8$ and $3/8$. Equal unmarked spectra therefore do not preserve this response.

If all permitted private readouts use a fixed mark $T(v)$, retain the joint reference distribution of $(\ell_z(v), T(v))$. A likelihood $L(o \mid z, T(v))$ then gives the conditional record probability

$$N_z(o) = \int e^r L(o \mid z, d)\eta_z(dr, dd), \qquad \eta_z = (\ell_z, T)_{\#}A. \tag{8.5}$$

After the record, the log-density coordinate changes by

$$r^+ = r + \log L(o \mid z, d) - \log N_z(o). \tag{8.6}$$

The public marginal is multiplied by $N_z(o)$ and normalized. Between records the same smooth conditional relaxation resumes, so the record's imprint on the private sector fades: the potential has no record term, and records act only at the moments they arrive. This gives an exact inherited record-and-update sequence for the declared protocols. If the marks identify every private point, however, the supposed compression may retain everything. Sufficiency is relative to what the future interaction can ask.

8.4 A smooth channel can erase a dependence in finite time

A separate four-agent example shows that exact independence can become available during the original smooth flow. Each belief and model law has a public bit, a private bit, and an internal gate bit. A fixed normalized local evaluator couples the public belief and model means. The gate means relax toward a fixed reference value. The private part of a peer channel depends on the source's private mean multiplied by a smooth function α of the population's gate means.

The function α has an exactly flat zero region. Its derivative is included in the Fisher equations. The gate is in effect an internal clock: it relaxes on its own, perturbed only by a term that a small private gain keeps dominated, and it enters the flat region at a finite time and stays there. Once inside, every peer's private cost in a receiver row is the same for all sources. It cancels from the Gibbs source comparison. The reduced energy and Fisher metric then split into a public part and an independent private-and-gate part. Both the finite-time erasure and the change of preferred source follow from a declared ordering of the private gain, the public peer strength, and the gate scale; the example does not show that the KL dynamics produces either without that ordering.

The public states start in two invariant sign regions, with two agents in each. No labels are assigned to groups by the equations. Initially, private similarities can select crossed pairs, each containing one member from each public region. After private erasure, the preferred sources change: each agent prefers the other member of its public region, and that preference persists. All cross-pair source weights remain positive. The two members of a preferred pair are distinct at every finite time, but this heterogeneity is transient: for a fixed temperature and a small enough public peer strength, their public states converge exponentially to each other, so each pair approaches a common state.

The interaction is not what produces this sequence. If every peer cost is deleted, the same rule for choosing preferred sources reads the crossed pairs first and the regional pairs afterward, because the private advantage and the public sign regions are supplied by the initial states; only the moment of the switch moves, since the gates then close more slowly. The example is therefore a single-stage selection with exact inheritance, not a formation.

For each pair, retain the empirical measure of its two paired public belief and model means. The original public vector field gives its exact evolution, and fresh public responses are computed by selecting and sampling a child from that measure. The private and gate laws can be forgotten after erasure, because they form a factor of the dynamics that has decoupled and that no admitted response reads. This is a noninjective reduction, but the empirical measure keeps every public child state, up to its label, and it is not itself a law pair of the children's type.

The example is an existence construction with an engineered channel, a product preparation on finite alphabets, and a specified class of boundary responses. It adds no fusion penalty and imposes no synchronization. It does not show that arbitrary smooth channels acquire a sufficient interface, that the distinct public child states admit a smaller aggregate description, or that the two resulting groups form another parent. Those are the remaining formation obligations.

Chapter 9

A collective trajectory is not a complete interface

Everything in this chapter is a statement about the reciprocal law potential, the cooperative consensus potential whose Fisher flow moves each source to reduce the costs of the receivers that attend to it. It is not a statement about the free energy of any agent's generative model. Within that potential there is a way to compress distinct public laws under the smooth interaction. It uses their collective motion. Near a common stable reference, differences between agents decay faster than their shared evolution. One can then assign each fine configuration a virtual collective state whose future follows exactly the same reduced trajectory. The fine agents need not coincide at any finite time, but they all approach the reference, so what is compressed is the decaying part of the approach. The construction is the asymptotic phase familiar from the theory of isochrons and isostables, carried out in charts of bounded log densities.

For an explicit general-law construction, take probability references A, B with balanced binary features φ, ψ, and the fixed normalized evaluator

$$G(m, dk) = \frac{e^{J\varphi(k)\psi(m)}}{\cosh J} A(dk), \qquad |J| < 1. \tag{9.1}$$

The bound $|J| < 1$ is exactly the condition for the common reference to be stable. The other sample coordinates can be arbitrary. Write $q_i = e^{u_i} A / \mathbb{E}_A e^{u_i}$ and $s_i = e^{v_i} B / \mathbb{E}_B e^{v_i}$, with bounded reference-centered log densities. No finite-dimensional family is imposed. Use identity channels, uniform nonself source priors, and the Gibbs-reduced reciprocal potential.

Set $z_i = (u_i, v_i)$. On the virtual diagonal the common law follows

$$\dot{u} = -u + J\varphi \mathbb{E}_{s_v} \psi, \qquad \dot{v} = -v + J\psi \mathbb{E}_{q_u} \varphi. \tag{9.2}$$

At the reference state, the linearized fine dynamics is

$$\dot{z}_i = \mathcal{A} z_i - \frac{2N}{N-1}(z_i - \bar{z}), \qquad \bar{z} = \frac{1}{N} \sum_i z_i. \tag{9.3}$$

Here $\mathcal{A}$ is the linearization of eq. (9.2). Its slowest common decay rate is $1 - |J|$, whereas the slowest difference rate is $1 + 2N/(N-1) - |J|$. The factor two in the peer term includes both receiving and being used as a source. This gap is what the interaction contributes. If the peer costs are switched off, each agent relaxes on its own with the common rates, the differences decay no faster than the average, and the construction below loses its foundation.

The arithmetic average of the log densities is not itself an exact closed state. Away from the diagonal, its equation contains a correction quadratic in the child differences. That correction decays fast enough to define a nonlinear phase. Let Φ_t be the fine flow and ϕ_t the flow of eq. (9.2). The exact parent coordinate is

$$\Pi(x) = \lim_{t\to\infty} \phi_{-t}\big(\overline{\Phi_t x}\big), \qquad \Pi(\Phi_t x) = \phi_t(\Pi x). \tag{9.4}$$

The backward common flow transports the late collective state back to the initial time. The fast decay of child differences makes the limit converge, and the limit is the same whichever smooth symmetric average is used. Its local fibers contain continuously many distinct child configurations, so this construction discards public degrees of freedom rather than merely their labels. The parent evolves exactly once its initial phase is known. The phase can be computed to exponential accuracy by integrating forward for a time t_0, averaging, and integrating the common flow backward for the same time. It is also the only choice: any Lipschitz parent map that is the identity on equal children and intertwines the two flows equals Π.

The response of the group is a separate question. For two binary agents with equal belief and model means within each agent, put $x_1 = m + a$ and $x_2 = m - a$. Their phase-law mean is

$$\bar{p} = m + \frac{3(2-J)}{2(5-J)} m a^2 + O(|m|a^2(|m| + |a|)^2). \tag{9.5}$$

A uniform choice of child followed by one bit sample has mean m. The diagonal configuration with both child means equal to $\bar{p}$ has the same phase parent, but its corresponding response has mean $\bar{p}$. For small $m = a > 0$, these means differ. The parent predicts its own inherited trajectory while failing to reproduce this ordinary group response. Sampling the phase law would change the interface. Because the phase is unique, no exact single-law parent of this kind can do better, although the defect is small: it is second order in the child differences.

An unrestricted identity-channel Gibbs receiver can distinguish even more. For child densities f_j and positive weights w_j, its block normalizer is

$$Z(r) = \sum_j w_j e^{-D_{\mathrm{KL}}(r\lambda \| f_j \lambda)/T}. \tag{9.6}$$

Knowing this response for all nearby receiver laws determines every distinct exposed child density and its total weight. To see the mechanism, cancel the known receiver entropy. The remaining function is a finite sum of exponentials of the linear probes $\int r \log f_j$. A direction that separates these probes gives a Vandermonde system, which recovers their coefficients. Distinct child laws therefore cannot be forgotten while retaining this entire interface. A fixed comparison channel changes the conclusion in a precise way: the receiver then sees only the channelled laws, and children with equal channelled laws are indistinguishable to it. For a deterministic coarse-graining channel even the force that the receiver exerts on its sources depends only on the channelled laws.

Restricting the possible receivers may permit compression. It must be accompanied by an exact check of the incoming forces and of repeated-response memory. For a second formation step, joining parents must also respect the information already discarded. Measured against the formation standard, the collective phase is a retention of a supplied block rather than a formation: nothing is

selected, the ordinary sampling response is not inherited, and the fine differences it compresses decay. It supplies a continuous autonomous reduction without an erasure gate, but not the recursive interface properties or a dynamically selected hierarchy.

Chapter 10

A Parent That Keeps Its Fresh-Sample Responses

A parent can now be constructed that preserves evolution and the fresh uniform-descendant response interface, through any finite number of supplied binary levels. The dynamics is the reciprocal law potential with its Fisher flow: the original smooth identity-channel KL interaction, with an evaluator that couples belief and model. The children keep distinct public laws. The construction works because a particular class of public patterns is invisible to fresh uniform-descendant sampling, and remains invisible under the inherited evolution. The tree is supplied, and nothing here shows the dynamics selecting it.

10.1 Four children and three public signs

Take a general base variable r with reference law A, and three independent public angles. Each angle determines a balanced sign, for example

$$\sigma_L = \operatorname{sign}\cos(\theta_L - \alpha_L(r)), \tag{10.1}$$

and similarly σ_R, σ_0 for the right node and the root. The phase $\alpha_L(r)$ can be any measurable function. Rotating it changes which actual public samples carry positive or negative contrast. A fair coin with a base-dependent orientation would serve equally well for every exact statement below; the angles are needed only to rotate a phase continuously.

Choose positive profiles $m_+(r), m_-(r)$ and contrast magnitudes $0 < a_+(r), a_-(r) < 1$, with $\int (m_+ + m_-) dA/2 = 1$. Four child densities are

$$\begin{aligned} q_{L\pm} &= m_{\sigma_0}(1 \pm a_{\sigma_0}\sigma_L), \\ q_{R\pm} &= m_{-\sigma_0}(1 \pm a_{-\sigma_0}\sigma_R). \end{aligned} \tag{10.2}$$

They are laws on the full base-and-angle space. There is a separate copy for the model laws. A fixed joint reference $\rho(r, w)$ couples the belief and model bases and is independent of the angles. Every child still compares its full law with every other child through identity channels.

If a response freshly chooses $L+$ or $L-$ with equal probability and then samples that child, its density is m_{σ_0}. The sign σ_L cancels pointwise. Its phase therefore changes the two individual public laws without changing this pair response. The pair nevertheless retains $a_{\sigma_0}^2$: its magnitude influences the future mixture through the KL forces.

The two first parents have response densities m_{σ_0} and $m_{-\sigma_0}$. They are distinct when $m_+ \neq m_-$. Choosing one parent uniformly cancels σ_0 and gives the full-coordinate response density

$$\bar{m}(r) = \frac{m_+(r) + m_-(r)}{2}. \tag{10.3}$$

The root also retains the unordered joint pair

$$\left\{(m_+(r), a_+(r)^2), (m_-(r), a_-(r)^2)\right\}. \tag{10.4}$$

It can discard the phase of σ_0 while retaining the magnitudes that determine its future. A density value and its contrast must remain paired.

10.2 Why the original dynamics closes

For a tree of depth d, assign a balanced public sign to every internal node. A leaf reads one sign on each step of its path. Write the resulting bit string as $u \in \{-1, 1\}^d$, and let its density be $f(r, u)$. This profile is an arbitrary positive bounded function; it is not required to remain a product of independent factors.

Two leaves that separate immediately after their first k common edges have the same first k bits and opposite next bits. Their later signs come from separate branches. Define P_k to keep the shared bits, flip the next bit, and average the remaining bits. The exact original peer cost is

$$D_k(f) = \int f[\log f - P_k \log f] dA dU_d, \tag{10.5}$$

where U_d is the uniform bit law. The source classes have evolving total weights b_k. Differentiating the complete energy gives

$$h = 2\log f + V - \sum_k b_k \left[P_k \log f + \frac{P_k f}{f}\right], \qquad \dot{f} = -f(h - \mathbb{E}_f h). \tag{10.6}$$

Here $V(r) = -\int g_0(w) \log \rho(r, w) dB(w)$ carries the other sector's base marginal. The ratio $P_k f / f$ comes from being a source in other agents' costs. Omitting it would give a different flow.

All quantities in eq. (10.6) use the profile values and their nested grouping. A branch exchange preserves them. At every physical sample point, the source leaves in each class run through every possible suffix bit string, so each leaf's original force equals the profile force eq. (10.6) evaluated at that leaf's own path bits. Nothing is averaged over receivers. Both the original energy and the original Fisher metric carry the factor 2^d; it cancels in the inherited equation. There is no inserted coarse timescale.

The reason this works is symmetry. The fine energy and the Fisher metric do not change when the public angles are relabeled in any way that keeps the base fixed and preserves the angle laws, or when agents are permuted. The family above is exactly the set of configurations left fixed by those relabelings that are compensated by a tree automorphism, so the flow cannot leave it. The phases, and the order of the two branches at each node, are therefore invisible to the energy, the metric, and the admitted records. This is the classical principle of symmetric criticality applied to a gradient

flow. What symmetry does not provide is the explicit cost D_k, its incoming term, and the formula for the fresh response.

The parent therefore keeps an unordered tree of profile values. Joining siblings places their two retained trees inside one unordered pair. Subsequent joins can be evaluated in the same way, and the result agrees with retaining the entire subtree directly. The grouping must remain: a flat multiset would forget which profiles enter each D_k. The parent state can still grow with the number of children.

This joining changes the description of one already coupled fine system. It does not switch on new edges or prove that the agents have selected that tree. Original uniform source rows are allowed; their class totals evolve under the original row equations.

10.3 What the inherited interface permits

At every level, choose a descendant freshly and uniformly, draw its full original sample, and conceal its label. The retained subtree mean gives the exact response density. Choosing a parent and then a descendant gives the same response as direct descendant selection. Repeat with fresh labels and fresh samples, using independent labels for belief and model responses.

Feedback also closes for record likelihoods that depend only on the base and are applied to every agent. In the reciprocal potential a record is an assimilation event: it updates the laws, and the potential itself is unchanged. Every child has the same base marginal $f_0(r)$, so a record with likelihood $L_e(o \mid r)$ has the common normalization $Z_o = \int f_0 L_e dA$. Its exact component update is

$$f^+(r,u) = \frac{L_e(o \mid r) f(r,u)}{Z_o}. \tag{10.7}$$

Every value at that base point receives the same multiplier, which commutes with every tree retention. The next experiment may depend on the previous record history. Induction then preserves the full finite history law, with the original continuous dynamics between updates. These admitted records see only the root marginal f_0. A record that reads a public sign, or a record delivered to only one subtree, multiplies the two branches by different factors and exposes the phase that the parent discarded.

Reusing one hidden child for two samples changes the response. For a pair $q_\pm = p(1 \pm a\sigma)$ it gives

$$p(\xi)p(\xi')[1 + a(\xi)a(\xi')\sigma(\xi)\sigma(\xi')]. \tag{10.8}$$

The cross term reads the discarded public pattern. Unrestricted external probes can also expose it. Exact inheritance is a statement about the declared interface, not every experiment one could perform on the children.

The discarded contrast is also transient. If the profile is close to one that does not depend on the bits, then to first order each contrast mode decays like $e^{-\lambda t}$ with $\lambda \geq 1$, and at the uniform source rows every mode decays at the common rate $3 + 2/(N-1)$. The same reciprocal flow that preserves the compression drives the distinctions it hides toward zero.

Finally, a phase rotation changes the public laws continuously in total variation: the angular region whose sign changes is proportional to the rotation angle. The corresponding squared Hellinger distance has the same linear dependence, so the Fisher–Rao distance grows like the square root of the angle. A phase path is therefore Hölder of order one half and has infinite Fisher–Rao length; it has no finite Fisher tangent. The original time evolution remains smooth on each fixed-partition profile chart. This construction establishes recursive boundary inheritance on a prepared family. Chapter 11 gives a different prepared family with selected membership and inherited Fisher geometry. It uses moment-dependent channels and an extensive interface; it does not establish selection of the identity-channel profile tree used here.

Chapter 11

Two-Stage Selection on a Wishart Family

This chapter concerns the reciprocal law potential, the direct interaction potential between agent laws whose Fisher gradient flow moves each source toward the receivers that attend to it. It is neither the global free energy nor the per-agent free energy of the dynamics chapter, and nothing here is transferred to those objectives. Four distinct agents are grouped into two parents and then into a parent of those parents, while each parent keeps the original smooth KL/Fisher dynamics, a declared response interface, and the collective connection of its children.

The example is best read as exact inheritance along a prepared ordering. The rule that groups the agents reads their belief shapes, and every belief shape relaxes as $\dot{\alpha}_i = -2(\alpha_i - \alpha_0)$, with no trace of the coupling, of the precisions, of the model sector, or of any other agent. The grouping is therefore the centroid dendrogram of the initial shapes, read on a clock whose origin the threshold sets, and the same grouping occurs when the agents do not interact at all. Under the formation standard of the inheritance chapter this fails interaction necessity, so the construction is a two-stage selection with exact inheritance, not a recursive formation. The architecture assumes common matrix precisions, a constant model-independent evaluator, a particular moment-dependent interaction, fresh sum responses, and a fixed membership threshold. All block-sum descriptions already exist mathematically at the initial time, and they are exact for every partition; the threshold selects when one becomes operative, and the dynamics does not first create the redundancy that permits it. The initial class is open within this statistical architecture.

11.1 Random matrices as represented contents

Let both represented-content spaces be $\mathsf{K} = \mathsf{M} = \mathrm{SPD}(2)$, the cone of real symmetric positive-definite two-by-two matrices. A sample k or m is a matrix content. It is distinct from a contextual coordinate and need not be the covariance of another random variable. Use the Wishart density

$$\begin{aligned} W_2(\nu, \mathsf{P})(dk) &= \frac{\det(\mathsf{P})^{\nu}}{\Gamma_2(\nu)} \det(k)^{\nu - 3/2} e^{-\operatorname{tr}(\mathsf{P}k)} dk, \\ \Gamma_2(\nu) &= \sqrt{\pi}\,\Gamma(\nu)\Gamma(\nu - \tfrac{1}{2}), \end{aligned} \tag{11.1}$$

where $\nu > 1/2$, $\mathsf{P} \succ 0$, and dk is Lebesgue measure on the three independent entries. The shape ν is real; a Gaussian sampling representation is not assumed. Γ_2 is the multivariate gamma function [23], and $\nu > 1/2$ is exactly the range of real shapes for which this density is a probability law [12]; Letac and Massam [20] discuss the harder noncentral case.

Normalization can be checked by writing $k_{11} = u$, $k_{12} = \sqrt{u}v$, and $k_{22} = w + v^2$, with $u, w > 0$. At $\mathsf{P} = I_2$, the determinant is uw, the trace is $u + w + v^2$, and the Jacobian is $\sqrt{u}$. Two gamma integrals and one scalar Gaussian integral give $\Gamma_2(\nu)$. Congruence by $\mathsf{P}^{-1/2}$ gives the general normalizer. The same calculation yields

$$\mathbb{E}e^{-\operatorname{tr}(Tk)} = \left(\frac{\det \mathsf{P}}{\det(\mathsf{P}+T)}\right)^{\nu}, \qquad \mathbb{E}k = \nu\mathsf{P}^{-1}, \qquad \mathsf{P}+T \succ 0. \tag{11.2}$$

Multiplying these transforms proves that independent matrices with common precision add their shapes. Transform uniqueness applies because the transforms are finite near $T = 0$. Thus $k_i \sim W_2(\alpha_i, \mathsf{P})$ independently implies $\sum_{i\in I} k_i \sim W_2(\sum_{i\in I}\alpha_i, \mathsf{P})$. This is an exact convolution identity for laws of matrices.

Take four agents with $q_i = W_2(\alpha_i, \mathsf{P})$, $s_i = W_2(\eta_i, \mathsf{Q})$, and product recognition $\zeta_i = q_i \otimes s_i$. Shapes exceed $1/2$. They may differ across agents and preparations, but are constant over the context base. They will evolve in inference time. The common precisions are preparation assumptions.

11.2 An interaction that moves every precision and no shape

Fix $\alpha_0, \eta_0 > 1/2$ and $\kappa > 0$. The proper local reference is the product of $W_2(\alpha_0, I_2)$ and $W_2(\eta_0, I_2)$; its normalized evaluator is the constant kernel $G_i(m, dk) = W_2(\alpha_0, I_2)(dk)$. The constancy matters. A belief evaluator whose precision depended on the model sample m would add the average of that precision over s_i to the force on q_i; since the model laws differ across agents, their precisions would separate and the sum identity above would no longer produce Wishart parents. With the constant evaluator the two sectors are uncoupled. Let $N = 4$ remain the primitive population count. Define

$$\begin{aligned} M_q &= \sum_i \mathbb{E}_{q_i} k, \qquad \Lambda_q = I_2 + \frac{\kappa}{N} M_q, \\ \mathcal{K}^q_{ij}[\boldsymbol{q}](k', dk) &= W_2(\alpha_0, \Lambda_q)(dk). \end{aligned} \tag{11.3}$$

The model sector exchanges (q, α_0, k) for (s, η_0, m). Every source, including self, is admitted with prior $\pi_{ij} = 1/N$; the self channel is deliberately not the identity, since an identity self channel would have zero cost and would tilt every row away from its prior. These normalized channels ignore the individual input sample k', but read the population's current moment. Each receiver's costs are identical across source labels, so exact Gibbs reduction at any positive source temperature gives the original prior row. The reduced belief potential is

$$\mathcal{E}_q = \sum_i \left[D_{\mathrm{KL}}(q_i \| W_2(\alpha_0, I_2)) + D_{\mathrm{KL}}(q_i \| W_2(\alpha_0, \Lambda_q)) \right]. \tag{11.4}$$

Changing one law changes every target in eq. (11.3). The moment-dependent peer terms are $\operatorname{tr}(\Lambda_q M_q) - N\alpha_0 \log\det\Lambda_q$. Their full differential is $\operatorname{tr}(C_q dM_q)$, where

$$C_q = \Lambda_q + \frac{\kappa}{N}(M_q - N\alpha_0\Lambda_q^{-1}) = I_2 + \frac{2\kappa}{N} M_q - \kappa\alpha_0\Lambda_q^{-1}. \tag{11.5}$$

The second term in the first expression comes from differentiating the channel. Freezing the target would remove it. Up to a constant, the density force and its ordinary unit-Fisher flow are

$$\begin{aligned} h_i(k) &= 2\log q_i(k) - 2(\alpha_0 - \tfrac{3}{2})\log\det k + \operatorname{tr}[(I_2 + C_q)k], \\ \dot{q}_i &= -q_i(h_i - \mathbb{E}_{q_i} h_i), \\ \dot{\alpha}_i &= -2(\alpha_i - \alpha_0), \qquad \dot{\mathsf{P}} = -2\mathsf{P} + I_2 + C_q. \end{aligned} \tag{11.6}$$

The last two equations follow by substituting eq. (11.1) into the density equation and matching the coefficients of $\log\det k$ and k. They establish tangency of the unrestricted density field. No parameter projection or mobility factor is inserted. The model equations use $(\eta_i, \eta_0, \mathsf{Q}, C_s)$.

The matching shows where the interaction acts. Substituting $\log q_i = (\alpha_i - \frac{3}{2})\log\det k - \operatorname{tr}(\mathsf{P}k) + \text{const}$ turns the force into $2(\alpha_i - \alpha_0)\log\det k$ plus a term linear in k. The coupling κ and the other agents live only in the linear term, so they move the precision and never the shape. The precisions are genuinely coupled: a moment perturbation H gives $DC_q[H] = (2\kappa/N)H + (\kappa^2\alpha_0/N)\Lambda_q^{-1}H\Lambda_q^{-1}$, whose trace pairing with $H \neq 0$ is positive. The product evaluator supplies no coupling between sectors.

A general picture lies behind this. When every receiver sees the same evaluator and the same channel target, and the target reads the laws only through an additive moment, the force on each agent is $2\log q_i$ plus one function common to all agents. Every log-likelihood ratio between two agents then decays as e^{-2t} whatever the interaction, and in an exponential family every difference of natural parameters does too. The precision difference $\mathsf{P}_i(t) - \mathsf{P}_j(t) = e^{-2t}(\mathsf{P}_i(0) - \mathsf{P}_j(0))$ and the shape law are two instances. A grouping rule that compares agents with each other can only see a shrinking copy of the starting configuration. Making the interaction matter for grouping requires targets that depend on the receiver.

These equations remain regular for all finite times. Put $\bar{\alpha} = \sum_i \alpha_i/N$. Then $M_q = N\bar{\alpha}\mathsf{P}^{-1}$ and each precision eigenvalue satisfies

$$\dot{p} = f(p, \bar{\alpha}(t)) = -2p + 2 + \frac{2\kappa\bar{\alpha}(t)}{p} - \frac{\kappa\alpha_0 p}{p + \kappa\bar{\alpha}(t)}. \tag{11.7}$$

The mean shape relaxes to α_0 and stays above $1/2$. The right side is positive for sufficiently small p and negative for sufficiently large p, uniformly along this shape trajectory. Each eigenvalue therefore remains in a compact positive interval. Shapes also remain in compact intervals above $1/2$, ensuring finite KL values and square-integrable scores. Common precision is preserved, and starting with unequal precisions they never become equal at a finite time.

11.3 The same clock after joining

For a current block I, retain $n_I = |I|$, $A_I = \sum_{i\in I}\alpha_i$, $B_I = \sum_{i\in I}\eta_i$, and the common precisions. Fresh independent addition gives $q_I = W_2(A_I, \mathsf{P})$ and $s_I = W_2(B_I, \mathsf{Q})$. The parent references and evaluator are the convolutions of the children's references and evaluator outputs; for the evaluator this is a kernel from the parent's model sum only because the child evaluators ignore the model sample. Their shapes are $n_I\alpha_0, n_I\eta_0$; the parent peer target has belief shape $n_I\alpha_0$ and precision Λ_q. Its source prior is $\pi^c_{IJ} = n_J/N$, the sum of the original source probabilities over sender block J.

The total moment is still $M_q = \sum_I \mathbb{E}_{q_I} k_I$, and $N = \sum_I n_I$ does not become the number of current parents. The inherited original-form potential is

$$\mathcal{E}_q^c = \sum_I \left[D_{\mathrm{KL}}(q_I \| W_2(n_I \alpha_0, I_2)) + D_{\mathrm{KL}}(q_I \| W_2(n_I \alpha_0, \Lambda_q)) \right]. \tag{11.8}$$

Its target normalizers still contribute $-N\alpha_0 \log\det \Lambda_q$. The complete derivative is consequently the same C_q, giving the unit-Fisher equations

$$\dot{A}_I = -2(A_I - n_I \alpha_0), \qquad \dot{B}_I = -2(B_I - n_I \eta_0), \qquad \dot{\mathsf{P}} = -2\mathsf{P} + I_2 + C_q, \tag{11.9}$$

and the exchanged equation for Q. These are exactly the summed fine shape equations and original precision flow, in the same time. Fine and parent energies need not be equal. Joining adds (n, A, B) and keeps the common precisions, so joining is associative and commutes with evolution. Independent addition also commutes with the evaluator and channel outputs. All of this holds for every partition, so inheritance alone does not single out the blocks that the rule below picks. Even the final single parent keeps its inherited nonidentity self channel and second KL term; deleting it would change the flow.

11.4 Two selected parents and an uncoupled lost allocation

Fix a threshold $\delta > 0$ and start with singletons whose belief shapes are pairwise farther than δ apart. A block's score is $b_I = A_I/n_I - \alpha_0$. It obeys $\dot{b}_I = -2b_I$; joining replaces two scores by their count-weighted average. Join current blocks when $|b_I - b_J|$ first reaches δ from above. Simultaneous threshold edges join their connected components. This is a declared, relabeling-equivariant rule applied to current states. Two shapes that start closer than δ would never be joined, because their distance never reaches δ from above; the separation assumption is part of the rule.

Because every score is the initial centroid times e^{-2t}, a pair of blocks reaches the threshold at $\frac{1}{2}\log(D/\delta)$, where D is the distance between their initial centroids. The first join is the closest pair. After it, the merged centroid lies between its two parts, and every other block lies outside that segment, so its new distance exceeds the merge distance. Heights therefore increase, nothing cascades, and the rule reproduces count-weighted centroid clustering of the initial shapes, the UPGMC method of numerical taxonomy [19, 29], with join k at time $\frac{1}{2}\log(D_k/\delta)$. It is not single linkage: for shapes $0, 1, 2.05, 3.5$ single linkage attaches the third point to the first pair, while the rule forms the two pairs first. Neither κ, nor the precisions, nor the model shapes enter, and changing δ only shifts every time by $-\frac{1}{2}\log\delta$. With the peer interaction switched off, the same events occur at the same times.

Sort the initial belief shapes as $u_1 < u_2 < u_3 < u_4$, and let d_1, d_2, d_3 be successive gaps. The lower pair and the upper pair are joined first, followed by their union, exactly when

$$\delta < \min(d_1, d_3) < d_2, \qquad \max(d_1, d_3) < d_2 + \tfrac{1}{2}\min(d_1, d_3), \tag{11.10}$$

up to ties. The first two times and the union time are

$$t_L = \tfrac{1}{2}\log(d_1/\delta), \qquad t_R = \tfrac{1}{2}\log(d_3/\delta),$$
$$t_2 = \tfrac{1}{2}\log\frac{d_2 + (d_1+d_3)/2}{\delta}. \tag{11.11}$$

To see why no other join intervenes, suppose the lower pair joins first. Its closest remaining singleton has distance $e^{-2t}(d_2 + d_1/2)$, larger than the upper pair's distance $e^{-2t}d_3$ by the second condition. Thus the upper pair joins next. The reversed ordering has the same argument. The resulting parent separation is $e^{-2t}[d_2 + (d_1+d_3)/2]$, the centroid distance rather than the nearest-neighbor gap d_2, so both first parents coexist for a positive interval. Each selected child-block distance remains below threshold afterward. All primitive shape differences remain nonzero at every finite time, but every law converges to one common limit: the parents are distinct at finite times and the population approaches consensus asymptotically.

For explicit values choose $\alpha_0 = \eta_0 = \kappa = 1$, $\delta = 0.1$, and initial belief shapes $(1, 1.2, 1.8, 2.1)$. The gaps are $(0.2, 0.6, 0.3)$ and the times are $\frac{1}{2}\log 2$, $\frac{1}{2}\log 3$, and $\frac{1}{2}\log(17/2)$; the first parents coexist for $\frac{1}{2}\log(17/6) \approx 0.52$. Model shapes can vary independently near $(0.8, 1.1, 1.4, 1.7)$. A small open neighborhood and all its relabelings preserve the strict inequalities. No binary tree was supplied; all additive block descriptions were available before any threshold crossing.

A formation time can reveal a belief-shape gap: at a join the two child scores differ by exactly δ, so the parent sum, the counts, and the time recover the children's belief shapes up to exchange. To exhibit information that remains lost even with all times recorded, fix the belief shapes and change two model shapes inside a joining pair by $(\eta_i, \eta_j) \mapsto (\eta_i + \epsilon, \eta_j - \epsilon)$. The parent model shape, population moments, subsequent precision flow, and all selection times stay unchanged. The individual public model laws change continuously, with a nonzero Fisher tangent because their shape scores contain the nonconstant statistic $\log\det m$. Distinct fixed belief labels exclude a mere permutation. At the second join, transfer model shape from one first-parent sum to the other while fixing the total. The final parent loses a further continuous distinction between its children. The lost allocation is, however, a spectator even before the join: each model shape relaxes on its own, the model precision sees only the total, the belief sector never reads the model laws, and the rule reads only belief shapes. The loss shows that the parent description is many-to-one, not that it discards something any agent responded to.

11.5 What a parent can be asked

At each query, draw fresh independent samples $k_i \sim q_i$, $m_i \sim s_i$ across agents and sectors. For every current block report $k_I = \sum_{i\in I} k_i$, $m_I = \sum_{i\in I} m_i$. At partition Π, the whole response vector has the conditional law

$$K_\Pi = \bigotimes_{I\in\Pi} \left[W_2(A_I, \mathsf{P}) \otimes W_2(B_I, \mathsf{Q})\right]. \tag{11.12}$$

Allow any normalized readout kernel on this vector. This includes overlapping unions of whole current blocks using the same batch of samples, so their correlations are retained. A controller may choose future readouts and finite query times from its carried record history and current retained state.

Condition on that history. Fine and parent waiting-time evolution agree by eq. (11.9), joins give the same retained state, and eq. (11.12) gives the same next-record kernel. Induction therefore gives the same finite joint history of records, actions, times, and retained states for each initial preparation. The argument is short for a reason: the interface sees only block sums, the retained state is a deterministic function of preparation and time, and records never feed back into the laws, so adaptivity only chooses when and what to read. Any expected cost or record entropy of an adaptive policy is therefore the same in both descriptions. Earlier records are carried forward; different preparations need not have identical record laws from time zero. Recording a response does not reset any belief. A query cutting through a retained block can reveal an unretained intersection shape. Reusing a hidden child's samples, applying child-specific Bayesian updates, or letting records update the laws requires another interface and another closure proof.

11.6 Prepared curvature that survives both joins and then fades

Use the contextual chart $(a, \chi) \in (1/5, 2/5) \times (-\chi_0, \chi_0)$, with $\chi_0 > 0$, and one $SO(2)$ acting by matrix conjugation on belief and model samples together. Let R_χ be rotation through angle χ, and prepare

$$\mathsf{P}(0; a, \chi) = R_\chi \operatorname{diag}(e^a, e^{-a}) R_\chi^T, \qquad \mathsf{Q}(0) = \operatorname{diag}(e^{3/10}, e^{-3/10}). \tag{11.13}$$

These coordinates describe how the laws depend on context, whereas k, m remain represented matrix samples. Context-constant shapes make all event times independent of context. The flow acts separately at each context, and nothing in it reads a connection: the connection below is computed from the evolved sections as a diagnostic. At fixed shape, a precision variation U has score and Fisher metric

$$\begin{aligned} \ell_U(k) &= \nu \operatorname{tr}(\mathsf{P}^{-1} U) - \operatorname{tr}(Uk), \\ g^F(U, V) &= \nu \operatorname{tr}(\mathsf{P}^{-1} U \mathsf{P}^{-1} V). \end{aligned} \tag{11.14}$$

The joint score for common contextual changes depends on the children only through their matrix sums. The same holds for common rotation scores. Addition therefore retains every contextual and orbit direction used here, with its exact Fisher inner products. Fine and parent infinitesimal stabilizers also agree. With unit child weights, the collective minimizing connection and residual contextual tensor are identical before and after a join.

Write $p_+ > p_-$ for the belief precision eigenvalues, and $\tilde{p}_+ > \tilde{p}_-$ for the model precision eigenvalues. The derivative f_p in eq. (11.7) is negative, and the scalar solution's derivative with respect to its initial value is $\exp(\int_0^t f_p du) > 0$ at finite time. Thus $\partial_a p_+ > 0$, $\partial_a p_- < 0$, and neither eigenvalue pair meets in finite time. For a block I, define its two rotational inertias and radial coefficient by

$$\begin{aligned} \mathcal{I}_1 &= 2A_I \frac{(p_+ - p_-)^2}{p_+ p_-}, & \mathcal{I}_2 &= 2B_I \frac{(\tilde{p}_+ - \tilde{p}_-)^2}{\tilde{p}_+ \tilde{p}_-}, \\ g_{aa} &= A_I \left[(\partial_a p_+ / p_+)^2 + (\partial_a p_- / p_-)^2 \right]. \end{aligned} \tag{11.15}$$

All three are positive, $\partial_a \mathcal{I}_1 > 0$, and $\mathcal{I}_2$ is independent of a. An angular connection correction ξ has Fisher cost $\mathcal{I}_1(d\chi + \xi)^2 + \mathcal{I}_2\xi^2$. Minimization, with the convention $D^{\mathcal{A}}p = dp + J_p\mathcal{A}$, gives

$$\begin{aligned} \mathcal{A}^F &= -\frac{\mathcal{I}_1}{\mathcal{I}_1 + \mathcal{I}_2} d\chi, & h^{\min} &= g_{aa}da^2 + \frac{\mathcal{I}_1\mathcal{I}_2}{\mathcal{I}_1 + \mathcal{I}_2} d\chi^2, \\ \mathcal{F} &= -\frac{(\partial_a \mathcal{I}_1)\mathcal{I}_2}{(\mathcal{I}_1 + \mathcal{I}_2)^2} da \wedge d\chi \neq 0. \end{aligned} \tag{11.16}$$

The scalar connection form represents the coefficient of the standard $SO(2)$ generator. Its curvature is its exterior derivative. The tensor has rank two at every finite time. Applying the same formulas to each block's fine children or retained parent gives equality through both joins. A passive frame rotation by ϑ sends $\mathcal{A}^F$ to $\mathcal{A}^F - d\vartheta$, with the compensating change in the section derivative. The fixed addition map contributes no moving-channel term to this geometric comparison. Physically, this is the gauge field of a two-part rotor whose belief part has an inertia that varies along the context while the model part does not; it is the mechanical connection of rotating bodies in a statistical setting [28].

Three qualifications keep this picture honest. First, the curvature exists because one rotation is declared to act on beliefs and models together. Nothing in the potential ties them; with separate rotations for the two sectors, the minimizing connection is $-d\chi$ on the belief factor and zero on the model factor, which is flat, and the tensor has rank one. Second, the contextual dependence was put into the initial section, and none of the formulas uses $\kappa > 0$: at zero coupling the same curvature appears. Third, the flow erases it. All contexts relax to the same isotropic precision $p_* I_2$, where p_* is the positive zero of $f(\cdot, \alpha_0)$, and eigenvalue splittings and their context derivatives shrink like $e^{-\lambda t}$ with $\lambda = -f_p(p_*, \alpha_0)$. For $\alpha_0 = \kappa = 1$, $p_* = \sqrt{2}$ and the tensor decays at rate $2\lambda = 12 - 4\sqrt{2} \approx 6.34$; by the root join it has lost more than three orders of magnitude. In the isotropic limit every rotation stabilizes the laws and the connection is undetermined. Keeping the geometry alive needs anisotropic, context-dependent local data in both sectors, for example a belief evaluator $W_2(\alpha_0, \Gamma(a, \chi))$ with Γ the prepared section and an anisotropic model reference; the precisions then relax to anisotropic fixed points and the curvature tends to a nonzero constant. The curvature is then a property of those declared data rather than of the interaction.

Context-constant shape allocations are substantive: an opposite contextual variation of two model shapes has positive fine Fisher length but zero derivative of their sum law. It would destroy the claimed tensor equality. Common precision, a constant model-independent evaluator, global moment channels, product recognition, and fresh extensive responses likewise delimit this construction. The threshold is constitutive, and selection does not alter the underlying fine flow. Chapter 12 removes the common-parameter preparation through a different selective architecture with native preferential routing and retained contextual profiles. Grouping that depends on the interaction, a derived interface or membership criterion, and unrestricted child probes remain separate problems. Neither a thermodynamic limit nor spacetime follows from these finite constructions.

Chapter 12

Preferential Groups from Open Initial Laws

This chapter works with the reciprocal law potential, the cooperative potential whose Fisher flow moves each source toward the receivers that attend to it. A prepared common-precision family makes exact matrix addition possible. A different construction starts from an open set of independently variable laws and lets native source preferences, read by a fixed threshold, determine which groups are selected. It uses two represented coordinates with different roles: a bit that communication transmits, and an angle whose relative constant offsets can be discarded. The parent retains enough angular profile information to preserve geometry. The selective architecture is supplied once; no interaction changes when a group is selected.

Three facts frame what the construction shows. The selection reads only the bits, while the compression is a symmetry of the angles that is available for every grouping at every time; the groups and times come from the bits, and the loss of angular information at a handoff comes from a declared response interface that becomes coarser when groups merge. In the regime of the proof, integrations indicate that switching off every peer interaction leaves the sequence of groups unchanged. Every group relation is temporary, and the same rule eventually merges everything. Measured against the formation standard, this is a two-stage selection with exact inheritance, not a recursive formation.

12.1 The transmitted bit and the retained angle

Take each belief and model possibility to be a pair $(z, \theta) \in \{-1, 1\} \times S^1$. Densities are relative to the uniform bit and circle probability measure. Every belief law q_i and model law s_i can have its own positive smooth bit-angle correlation. Recognition uses $q_i \otimes s_i$. Local reference laws and evaluators are uniform, source priors are fixed and uniform, all sources including self are admitted, and the temperature T is fixed and positive.

The comparison channel transmits z and generates a new uniform angle. It is used on every edge, the self edge included. If $\mu_i(z)$ is the binary marginal density, then

$$D_{\mathrm{KL}}(q_i \| \mu_j \otimes \mathrm{Haar}) = D_{\mathrm{KL}}(q_i \| \mu_i \otimes \mathrm{Haar}) + D_{\mathrm{KL}}(\mu_i \| \mu_j). \tag{12.1}$$

The first term depends on the receiver but not on the source. It cancels when the receiver normalizes its source probabilities:

$$\beta_{ij} = \frac{e^{-D_{\mathrm{KL}}(\mu_i \| \mu_j)/T}}{\sum_k e^{-D_{\mathrm{KL}}(\mu_i \| \mu_k)/T}}. \tag{12.2}$$

The model source row is the same expression using its own binary marginals. The cancellation needs the selective self channel: with an identity self channel the self cost would be zero while every peer cost kept the receiver term, and at low temperature the peers would be suppressed. In the language of the smooth-closure chapter, the bit is the public coordinate, the one the channels transmit, and the angle is private. The angle is absent from this selection equation but is still part of each agent's law, of the local force, and of an aggregate response defined below. Because every channel randomizes the angle, no agent ever holds a common angular frame with another; this is the familiar situation of parties without a shared reference frame.

Cancellation from a row is not cancellation from the variational potential. The complete belief force is

$$\begin{aligned} h_i^q &= 2\log q_i - \sum_j \beta_{ij}\log\mu_j - \sum_k \beta_{ki}\frac{\mu_k}{\mu_i}, \\ \dot q_i &= -q_i(h_i^q - \mathbb{E}_{q_i} h_i^q). \end{aligned} \tag{12.3}$$

One entropy comes from the local KL and the other from the normalized receiver row. The final sum is the incoming source derivative. The self ratio is one, but a self channel still replaces the angle, so its full divergence is generally nonzero. All these terms are required by the original smooth potential.

The formula is well defined on a positive smooth-law neighborhood. Logarithms and ratios are bounded there, and the force contains no angular derivative, so local solutions and their dependence on initial densities follow from an ordinary differential equation on a Banach chart. No Gaussian approximation or finite-dimensional moment closure is needed.

12.2 Why pairs appear before a larger group

The force splits exactly. Write $q_i(z,\theta) = \mu_i(z) f_i(\theta \mid z)$. The angular conditional at each bit value relaxes on its own toward the uniform law,

$$f_i(\theta \mid z, t) = \frac{f_i(\theta \mid z, 0)^{e^{-2t}}}{\int f_i(\psi \mid z, 0)^{e^{-2t}} d\psi/(2\pi)}, \tag{12.4}$$

and the binary means obey their identity-channel Fisher equations plus a small known forcing proportional to the difference of the two conditional entropies. For product laws the forcing vanishes. Thus an angular factor $e^{\kappa\cos(\theta-\chi)}/I_0(\kappa)$ keeps its phase and has concentration κe^{-2t}. The decay rate two is calculated from eq. (12.3); it is not a new slow mobility. Because the bit equations see angles only through rotation-invariant entropies, nothing that decides the groups can depend on an angle.

The binary routing mechanism is easiest to see at small amplitude. Write the mean as $x_i = h u_i$ and choose one temperature $T = h^2$. Without rescaling time, the limiting equation is

$$\dot u_i = -u_i + \sum_j (w_{ij} + w_{ji})(u_j - u_i), \qquad w_{ij} = \frac{e^{-(u_i-u_j)^2/2}}{\sum_k e^{-(u_i-u_k)^2/2}}. \tag{12.5}$$

The local term moves every mean toward zero. The peer terms act strongly on nearby means and weakly on well-separated means. Self mass remains in the denominator. This is the gradient flow of a Gaussian log-sum-exp energy, the same clustering mechanism as Gaussian mean shift and self-attention particle dynamics.

Place four reference coordinates at

$$-A-2, \quad -A+2, \quad A-2, \quad A+2 \tag{12.6}$$

with large finite A. Their internal gaps shrink while their centers are still far apart. Initially even a partner's native weight is at most e^{-8}, so the pairs do not begin as activated attention cliques. In the exact symmetric calculation, the half-distance a between pair centers and the half-gap b obey, while $a \geq 4, b \leq 2$,

$$-1.0014a \leq \dot{a} \leq -a, \qquad -3.03b \leq \dot{b} \leq -0.977b. \tag{12.7}$$

A partner weight crosses $1/10$ while exterior weights remain tiny. When the centers reach $a = 4$, the residual gap satisfies

$$b \leq 2(A/4)^{-0.975}. \tag{12.8}$$

It can therefore be made arbitrarily small without setting the children equal.

In the zero-gap comparison, the inherited source mass from one pair to the other is

$$B = \frac{e^{-2a^2}}{1+e^{-2a^2}}, \qquad \dot{a} = -a(1+4B). \tag{12.9}$$

It grows through $1/10$ at $a = \sqrt{\log 9/2}$. This crossing occurs while the original partner weights still dominate their complements. A small nonzero residual gap preserves the strict inequalities and the crossing. The second stage therefore happens when the supplied separation A has shrunk to order one, at a time close to $\log A$. The same formulas hold for two collapsed clusters of any common size, which suggests deeper hierarchies by repeating the scale separation; that extension is not proved.

Two pairs alone would leave no exterior at this second event. Add a third pair initially near a much larger finite coordinate R. Over the finite observation interval its center is close to Re^{-t}, and every force between it and the core is bounded by a constant times $(1+D)e^{-D^2/2}$, where D is their separation. Choosing R sufficiently large keeps every mass between this pair and the four-agent core below 10^{-3}. It forms internally but remains a genuine exterior while the two core parents join.

For a current partition, define the inherited row by averaging receivers and summing source columns:

$$B^q_{IJ} = \frac{1}{|I|}\sum_{i\in I}\sum_{j\in J}\beta_{ij}, \tag{12.10}$$

and similarly for the model sector. A proposed pair must satisfy, in both directions and sectors,

$$B^a_{IJ} > \frac{1}{10}, \qquad B^a_{IJ} - \sum_{K\notin\{I,J\}} B^a_{IK} > \frac{1}{20}. \tag{12.11}$$

Self loops are retained in normalization. The same condition selects primitive pairs and parent pairs. It has no competing partners at one vertex, because two partners could not each exceed the other by the positive contrast. At later joins, receiver averages use the inherited primitive counts. Neither temperature nor source data is refitted.

The limiting equation is a proof device. First choose finite A, then finite R, and finally one sufficiently small positive h. Smooth convergence transfers all strict finite-time events to the exact binary KL field, through one lemma that controls both trajectories and the event observables. The symmetry and product form are also proof devices: the exact split above transfers the event windows to a full open set of independent positive belief and model densities, with bit-angle correlations of size comparable to h. The initial class is open, not universal. Equal uniform laws, for example, do not produce the required contrast. The constants are not explicit. With the proof's choices the far pair sits near $u \approx 2 \times 10^7$, and the temperature implied by the written estimates is below 10^{-15}, or about 10^{-21} once the slower decay of distant means in the exact field is accounted for.

12.3 What switching off the interaction does

The formation standard asks whether a grouping would also occur with the interaction switched off. Deleting every peer cost leaves $\dot{u} = -u$, so $u(t) = e^{-t}u(0)$, and the rows become a Gaussian kernel of growing width applied to the initial positions. Integrations from the proof's initial data, at $A = 8$, 64, and 4096, give the same three pairs, the same parent join, and the same exterior under both flows; the events move by less than ten percent. These integrations use the limiting equation, not the exact field at the theorem's temperature, and they are not a theorem. What is proved is that peer forces contract the pairs faster than the null flow does. The groups themselves are fixed by the scale hierarchy written into the initial data.

The groups are also temporary. Every mean tends to zero, so every row tends to $1/6$. A first-stage contrast tends to $1/6 - 4/6 < 0$, and once only the merged core and the third pair remain, the contrast condition has no exterior and the two remaining rows tend to $2/6$ and $4/6$. The same rule therefore merges everything at a later finite time. The windows in the theorem are finite by construction.

12.4 What the parent actually forgets

An active rotation turns both laws of one agent by an angle in fixed sample coordinates. It leaves the agent's binary marginals, entropy, and evaluator unchanged, and the full dynamics is equivariant: a rotated initial law gives the rotated trajectory. The parent of I discards rotations whose angles sum to zero:

$$\mathsf{H}_I = \{(a_i)_{i\in I} : \sum_i a_i = 0 \pmod{2\pi}\}. \tag{12.12}$$

A two-child parent loses one continuous relative orientation. Joining two parents loses one further relative block orientation. These are changes of probability laws, not passive changes of coordinates. Whether they matter depends on who can see them. No agent, channel, row, or evaluator of the population distinguishes them; every one of these quantities is unchanged by independent rotations

of all belief and model angles. They are distinguished only by an external querier, declared with the construction, who reads angles in a fixed frame that no agent holds.

To represent the retained state by actual normalized laws, fix a contextual anchor c_*. Let ϑ_i be the phase of the belief first circular moment there and $\phi_I = |I|^{-1} \sum_i \vartheta_i$, using a local lifted-angle chart. Rotate the entire contextual section of child i by $\phi_I - \vartheta_i$ and take products of the rotated laws:

$$q_I(c) = \bigotimes_{i \in I} \tau_{\phi_I - \vartheta_i} q_i(c), \qquad s_I(c) = \bigotimes_{i \in I} \tau_{\phi_I - \vartheta_i} s_i(c). \tag{12.13}$$

This is a slice in the sense of the method of slices for symmetric evolution equations, using the first Fourier mode. The parent still contains all centered profiles and all changes of phase away from the anchor. It loses the constant relative offsets. Its state size grows with the number of children, and every child's bit marginal is kept, so the parent does not compress the public laws into a smaller collective state.

The equal-phase choice cannot be made globally: no rule that treats all children alike and turns with a common rotation can pick a phase from n phases, because rotating $(0, 2\pi/n, \dots)$ by $2\pi/n$ merely relabels it. A global representative exists if one child is singled out, and nothing observable depends on the choice.

For a subsequent join,

$$\phi_{I \cup J} = \frac{|I|\phi_I + |J|\phi_J}{|I| + |J|}. \tag{12.14}$$

The resulting primitive rotations equal those obtained by joining all children directly. Native routing is invariant under the discarded rotations, so even a complete record of the event times does not identify their offsets.

The parent has exact dynamics at the original time, but its moving representation must be differentiated. If ω_i is the current angular velocity of the first moment at the anchor, then

$$\partial_t \widetilde{q}_i = V_i^q(\widetilde{x}) - (\overline{\omega}_I - \omega_i) \partial_\theta \widetilde{q}_i. \tag{12.15}$$

Every quantity is determined by retained factors. The second term keeps the representative aligned as it evolves; it does not add a force to the fine system. Omitting it would generally change the parent trajectory.

An exact inherited vector field need not be the ordinary Fisher gradient of a new single KL model. For two phase inertias I_1, I_2, the quotient metric for total phase F is $dF^2/(I_1^{-1} + I_2^{-1})$, while the equal-phase representative has metric $(I_1 + I_2)dF^2/4$. These differ when the inertias differ. The fine velocity is orthogonal to the discarded rotations, so the parent loses its potential at the rate given by the quotient metric, and on finite-dimensional von Mises families the parent flow is exactly the gradient flow for that metric. The correction term in eq. (12.15) is the part of the representative's motion along the discarded rotations. The construction uses the derived flow in eq. (12.15) and retains the primitive interaction ports.

12.5 Which responses survive

Before any selection, declare that a query to a current block returns the bit vector and the sum modulo 2π of independent constituent angles. A model query uses the analogous product of model laws. The same rule applies at every block size. Native source choices and transmitted bits are also available, and future query choices may depend on the entire recorded history.

The sum of canonicalizing rotations is zero, so the response distribution calculated from eq. (12.13) is exactly the original one. It has a nontrivial angular signal: its first moment is the product of the constituent first moments. At the next join, bit vectors concatenate and angular sums add.

Equality of the waiting-time flow and the next joint response kernel proves equality of every finite adaptive history by conditioning successively on its records. This concerns the full joint history, not just its one-time marginals. Earlier finite sample records are carried forward. Positivity means that they do not exactly reconstruct the discarded continuous offsets. They do carry information about them: each angle sample of a singleton carries Fisher information equal to its rotation inertia, so many pre-merge samples estimate the offsets well. The loss is that no finite record set pins them down exactly. An exact archive of the child laws would remove the advertised loss.

Individual absolute-angle probes of a child inside a compressed block are not allowed by this interface after the merge. They change under a discarded relative rotation and cannot be recovered. The construction therefore preserves specified experiments, not every possible microscopic measurement.

12.6 Selection and inheritance are independent

Nothing in the last two sections used the selected groups. The rows and event times are unchanged by independent rotations; the slice, the moving flow, the response descent, and the geometric equality below hold for any grouping at any time, including at the start. The bits choose the groups and the times. The response rule decides that, from those times on, only angle sums are asked for. The orientation information lost at a handoff is exactly what the coarsened interface stops asking about. This is why the theorem is stated as a conjunction of selection, interface coarsening, and symmetry reduction.

12.7 Why the collective geometry remains

At any fixed time, the canonicalizing angle in eq. (12.13) is constant across evaluation context. The contextual derivative of a rotated law is therefore just the rotated original derivative. Rotation preserves Fisher inner products. Independent product scores have zero mean, so their cross terms vanish. The actual parent Fisher metric equals the fine sum on these contextual directions.

The common collective rotation acts on every primitive belief and model coordinate at once and commutes with the active rotations. This tie of belief and model angles to one group is a declared choice: if they were rotated independently, the minimizing connection below would be flat and the tensor would have rank one. Consequently the full cost

$$g(dp + J\xi, dp + J\xi) \tag{12.16}$$

is unchanged for every common angular velocity ξ. Its minimizer, minimized tensor, and local curvature are the same before and after either join. Nothing in the dynamics reads this connection; it describes the evolved sections.

This statement has nonzero examples throughout the same finite interval. Let belief angles have von Mises concentration κ and phase χ, and model angles have fixed positive concentration λ and phase zero. Use (κ, χ) as contexts and keep the reference binary trajectory independent of them. Define

$$\mathcal{I}(v) = v\frac{I_1(v)}{I_0(v)}, \qquad a_t = \mathcal{I}(\kappa e^{-2t}), \quad b_t = \mathcal{I}(\lambda e^{-2t}). \tag{12.17}$$

The function $\mathcal{I}$ is the rotation Fisher inertia. Integration by parts gives $\mathcal{I}(v) = \mathbb{E}_{p_v}[v^2 \sin^2\theta]$, and $\mathcal{I}'(v) > 0$ for $v > 0$. Minimizing the cost $a_t(d\chi + \xi)^2 + b_t\xi^2$ gives

$$A_t^F = -\frac{a_t}{a_t + b_t}d\chi, \qquad F_t^F = -\frac{b_t e^{-2t}\mathcal{I}'(\kappa e^{-2t})}{(a_t + b_t)^2}d\kappa \wedge d\chi. \tag{12.18}$$

For a block of size n, the two positive tensor components are

$$h_{\chi\chi} = n\frac{a_t b_t}{a_t + b_t}, \qquad h_{\kappa\kappa} = ne^{-4t}\operatorname{Var}_{p_{\kappa e^{-2t}}}(\cos\theta). \tag{12.19}$$

The tensor has rank two and the curvature is nonzero at every finite time. As time grows the tensor shrinks like e^{-4t}, while the curvature approaches $-2\lambda^2\kappa/(\kappa^2 + \lambda^2)^2$, which is not zero: what degenerates at long times is rank, not curvature. The curvature is present from the start and does not depend on which groups form; the retention keeps it because it is an isometry for every grouping. The geometry is supplied by this declared family. Repetition is the reference example, not a restriction on all admitted laws. To extend it through arbitrary nearby law values, tilt each belief by $e^{(\kappa-\kappa_*)\cos(\theta-\vartheta_i)}$, normalize, and rotate by χ; keep its model law constant over context. The tilt follows the law's own phase, so an active rotation rotates the whole assigned section. It cannot reveal an erased offset through a fixed external template. Smooth dependence then preserves the positive geometric bounds on a full open neighborhood.

The handoff is selected at one fixed anchor and applied to the entire section at the same time. If different contexts use different event times $\tau(c)$, differentiating their states adds $V d\tau$ and gives a different comparison. The phase roots are also local: a global block phase has a finite root ambiguity. Neither issue can be removed by treating it as notation.

Appendix A

Notation in Physical Terms

Symbol	Meaning and role
k, m	A represented state possibility and a model presentation. Neither symbol by itself denotes a probability law.
q_i, s_i	The agent's belief about represented contents (its perception) and its belief over generative models. They can evolve in coupled equations.
ζ_i	The joint law of represented state and model contents; $q_i \otimes s_i$ is a declared product restriction.
g^F, ℓ_v	Fisher distinguishability on a regular statistical family and the score describing one infinitesimal change.
$D^\omega\zeta, h^\omega_\zeta$	Statistical change after contextual transport is accounted for, and the resulting tensor on the context base.
$A^F, \mathfrak{k}$	The collectively minimizing connection class and the metric measuring common infinitesimal frame motions.
$G_i(m, dk)$	An evaluator assigning state predictions to each model presentation. Averaging gives $s_i G_i$.
r_i, ℓ_i, P_i	Prior over presentations, record likelihood, and the normalized generative model $r_i G_i \ell_i$. A model law predicts records only through G_i and ℓ_i.
Species	Agents that share a generative model up to a change of frame. They may still hold different beliefs.
Three objectives	A global free energy of one normalized joint over all agents; per-agent free energies in which peers' laws are data; the reciprocal law potential, which moves sources toward their receivers and is not a free energy of any fixed model with the agent laws as recognition laws.
$\mathcal{F}_i$	Free energy of agent i's own generative model. In the per-agent picture its own evaluator and its peers' channelled laws compete as sources of prior.
$x \in \mathsf{S}$	A complete configuration of agents, including their laws and the memory and interfaces read by the dynamics.
$\rho_i : \mathsf{S}_i \to \mathsf{O}_i$	A readout of an internal distinction. Its value is a record.
Γ_{ij}	A normalized interaction transition on complete configurations. It induces record probabilities through the readout.
U_i^a	An invertible frame map for sector $a = q, s$. Changing frame changes the description.
T_j^a, O_i^a	Sending and receiving interfaces. Their product can discard directions.
L_{ij}^a, Θ_{ij}^a	A reference generator and its framed exponential interaction. The generator may be singular while its exponential is invertible.
β_{ij}, γ_{ij}	Belief-sector and model-sector source weights in normalized rows.
$\mathcal{E}$	The reciprocal law potential on agent laws. Its gradient dynamics requires a metric, and every edge pushes on both its receiver and its source.
$\mathcal{F}$	Variational free energy for a specified normalized probabilistic model and conditioned record.
$Q_x C$	A law of compressed represented contents. It does not specify a parent update.
$R(x), F^c$	A complete retained state and its inherited update. They must satisfy $R \circ F = F^c \circ R$, with $\circ$ composing maps in function order.
Φ^0	The null flow: the same construction with every peer cost deleted. It is the control experiment for selection.

Symbol	Meaning and role
P, P^0	Partition sequences selected along the flow and along the null flow, without event times.
(F5)	Interaction necessity: $P \neq P^0$ on an open set of admitted initial data. A shift of event times alone does not count.
Two-stage selection	Short for two-stage selection with exact inheritance: selection, inheritance, and a second selection among parents, with at least one of compression, interaction necessity, or counterfactual dependence of the second stage failing. It is not a formation, and the failed clauses are stated.
$W_2(\nu, \mathsf{P})$	A Wishart law of positive-definite matrix contents, with real shape $\nu > 1/2$ and precision P. It is not a Gaussian closure.
$\alpha_i, \eta_i; A_I, B_I, n_I$	Individual belief and model shapes, their block sums, and the number of primitive children retained by a parent.
B^q_{IJ}, B^s_{IJ}	Native source masses between current blocks, averaged over receivers with primitive-count weights.
H_I, ϕ_I	Relative constant angular offsets discarded by a block, and its retained local mean phase at the contextual anchor.
$\mathcal{C}, B, F$	The context base, a connection one-form, and its curvature. These require a rule for relating neighboring contexts. The symbol F also denotes a discrete update in the inheritance chapter.

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